

- 19.** The terminal potential difference of a cell is 19 V when a current of 1.0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3.0 A. Find the emf and internal resistance of the cell.

2

Finding the emf

1

Finding the internal resistance

1



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2

|                                   |               |
|-----------------------------------|---------------|
| $V = E - Ir$                      | $\frac{1}{2}$ |
| $19 = E - 1.r$ _____ (i)          |               |
| $17 = E - 3.r$ _____ (ii)         | $\frac{1}{2}$ |
| $2r = 2 \Rightarrow r = 1 \Omega$ | $\frac{1}{2}$ |
| $E = 19 + 1 = 20V$                | $\frac{1}{2}$ |

Finding the emf

1

Finding the internal resistance

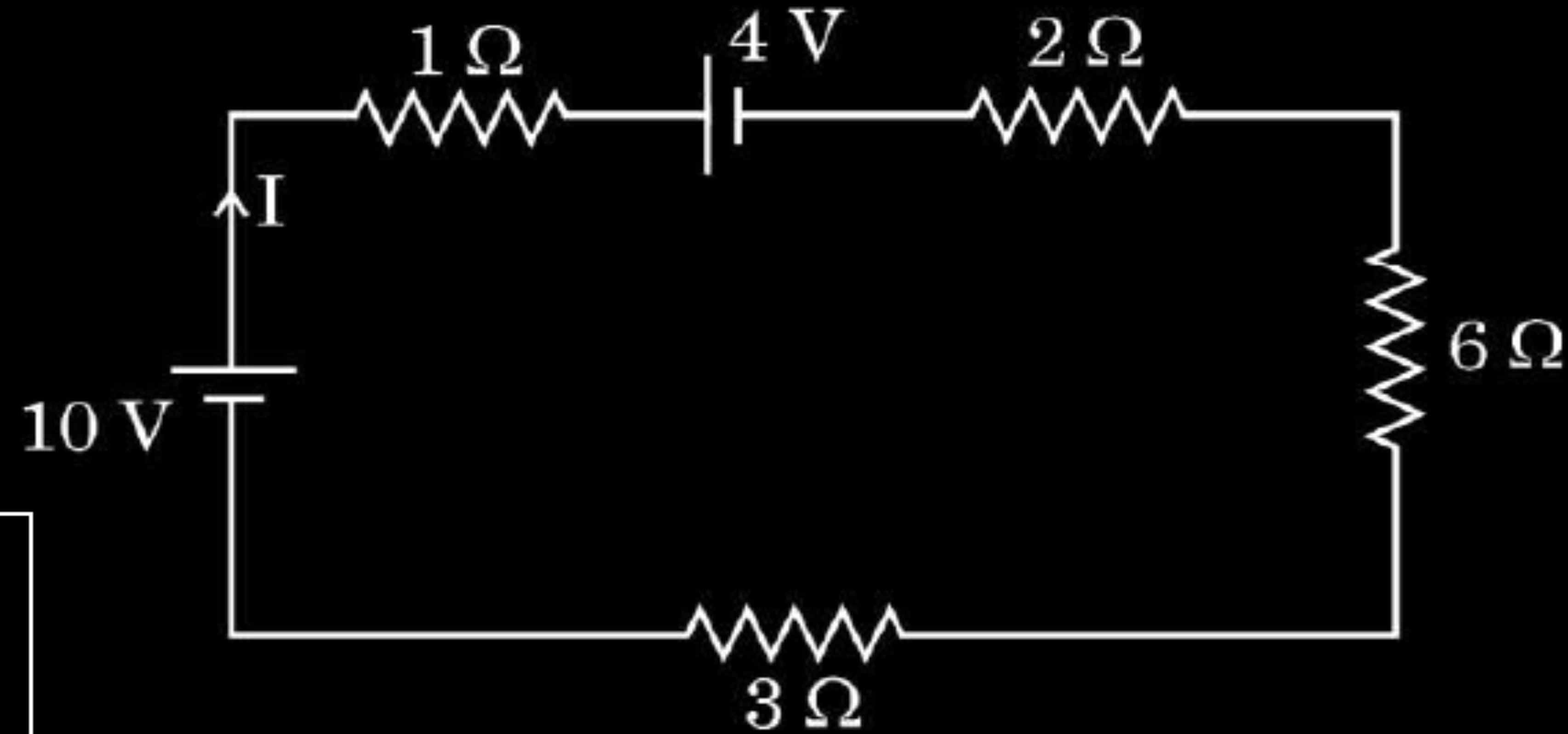
1



**OR**

- (b) (i) Define electrical conductivity of a wire. Give its SI unit.
- (ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries ?
- (iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.

5



|                                                                                  |     |
|----------------------------------------------------------------------------------|-----|
| (i) Definition & S.I. Unit                                                       | 1½  |
| (ii) Explanation of internal resistance for low voltage and high voltage battery | 1+1 |
| (iii) Total energy                                                               | 1½  |



|                                                                                                                                                                                                                                                                                                          |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <p><b>(i) Definition :</b> Electrical conductivity is defined as the measure of a material's ability to carry a current through it.</p> <p><b>Alternatively:</b> It is the reciprocal of the resistivity.</p> <p><b>Alternatively:</b> It is defined as the current density per unit electric field.</p> | 1             |
| <p><b>S.I. Unit:</b> <math>(\text{ohm})^{-1}\text{-m}^{-1}</math> or <math>\text{S-m}^{-1}</math></p>                                                                                                                                                                                                    | $\frac{1}{2}$ |
| <p><b>(ii) Low voltage Battery-</b><br/>Internal resistance should be low.</p>                                                                                                                                                                                                                           | 1             |
| <p><b>High voltage Battery –</b><br/>Internal resistance should be high.</p>                                                                                                                                                                                                                             | 1             |
| <p><b>(iii) Applying Kirchhoff's loop rule</b></p> $10 - I \times 1 - 4 - 2I - 6I - 3I = 0$ $12I = 6 \Rightarrow I = 0.5 \text{ A}$ <p>Heat energy <math>H = I^2 R t</math></p> $H = 0.25 \times 12 \times 60 = 180 \text{ J}$                                                                           | $\frac{1}{2}$ |
|                                                                                                                                                                                                                                                                                                          | $\frac{1}{2}$ |



- 22.** A cell of emf  $E$  and internal resistance  $r$  is connected to a variable resistance  $R$ . Draw plots showing the variation of (a) terminal voltage  $V$  with  $R$ , and (b)  $V$  with current  $I$ , in the circuit.

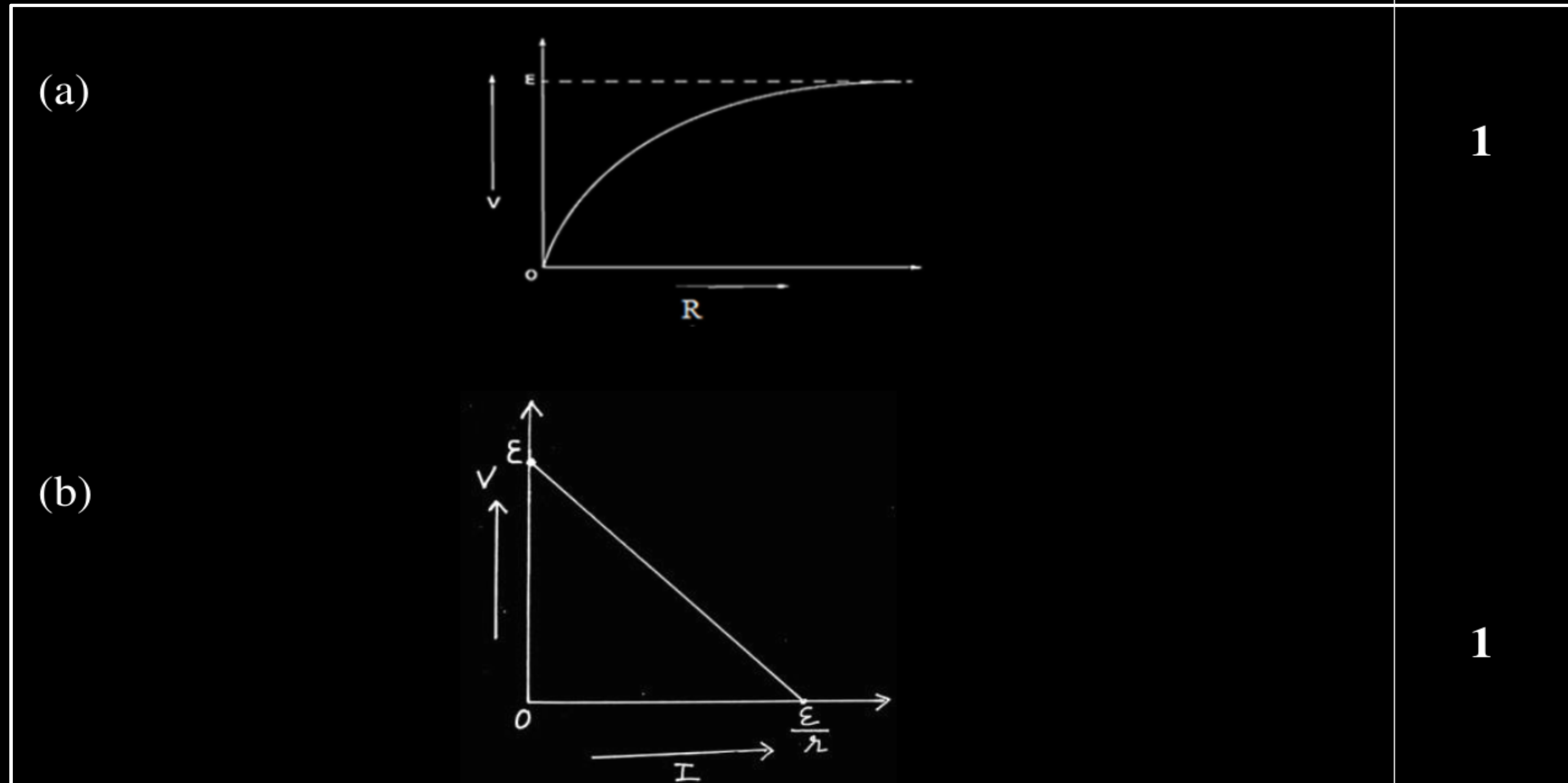
2

|                               |   |
|-------------------------------|---|
| Plots of                      |   |
| (a) Terminal voltage with $R$ | 1 |
| (b) $V$ with current $I$      | 1 |



22. A cell of emf  $E$  and internal resistance  $r$  is connected to a variable resistance  $R$ . Draw plots showing the variation of (a) terminal voltage  $V$  with  $R$ , and (b)  $V$  with current  $I$ , in the circuit.

2



Plots of

(a) Terminal voltage with  $R$

1

(b)  $V$  with current  $I$

1



- 25.** Two identical cells, each of emf  $E$  and internal resistance  $r$ , are connected with a load resistance  $R$ , first in series and then in parallel. Obtain the condition under which the current through  $R$  is same in both cases. 2

Calculation of current for cells in series and parallel  $\frac{1}{2} + \frac{1}{2}$   
Obtaining the condition when current is the same in both the cases  
1



**In series,**

$$\varepsilon_{\text{eq}} = \varepsilon + \varepsilon = 2\varepsilon$$

$$I_s = \frac{2\varepsilon}{2r + R}$$

**In parallel,**

$$\varepsilon_{\text{eq}} = \varepsilon$$

$$r_{\text{eq}} = \frac{r}{2}$$

$$I_p = \frac{\varepsilon}{\frac{r}{2} + R}$$

$$\therefore I_p = I_s$$

$$\frac{2\varepsilon}{2r + R} = \frac{\varepsilon}{\frac{r}{2} + R}$$

$$\frac{2}{2r + R} = \frac{1}{\frac{r}{2} + R}$$

$$2\frac{r}{2} + 2R = 2r + R$$

$$r = R$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\mathbf{1}$$



**20.** Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell ?

2

Two differences  
Important precaution

$\frac{1}{2} + \frac{1}{2}$   
1



**20.** Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell ?

2

|                                                                                                                                                                                                                                                                                                                                   |                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <p><b>Two differences</b></p> <ol style="list-style-type: none"> <li>1. The potential difference across the electrodes in open circuit is e.m.f. ( <math>\varepsilon</math> ) and in closed circuit is terminal potential difference (V).</li> <li>2. V depends on r and <math>\varepsilon</math> is independent of r.</li> </ol> | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |
| <p><b>Precaution-</b></p> <ol style="list-style-type: none"> <li>1. Some external resistance should be connected to cell in series.</li> <li>2. Short circuiting should be avoided.</li> </ol>                                                                                                                                    | <p>1</p>                                                        |

Two differences  
Important precaution

$\frac{1}{2} + \frac{1}{2}$   
1



1. A battery of emf  $E$  and internal resistance  $r$  is connected to an external circuit. The potential drop within the battery is proportional to :
- (a) current in the circuit
  - (b) total resistance of the circuit
  - (c) emf of the battery
  - (d) power dissipated in the circuit



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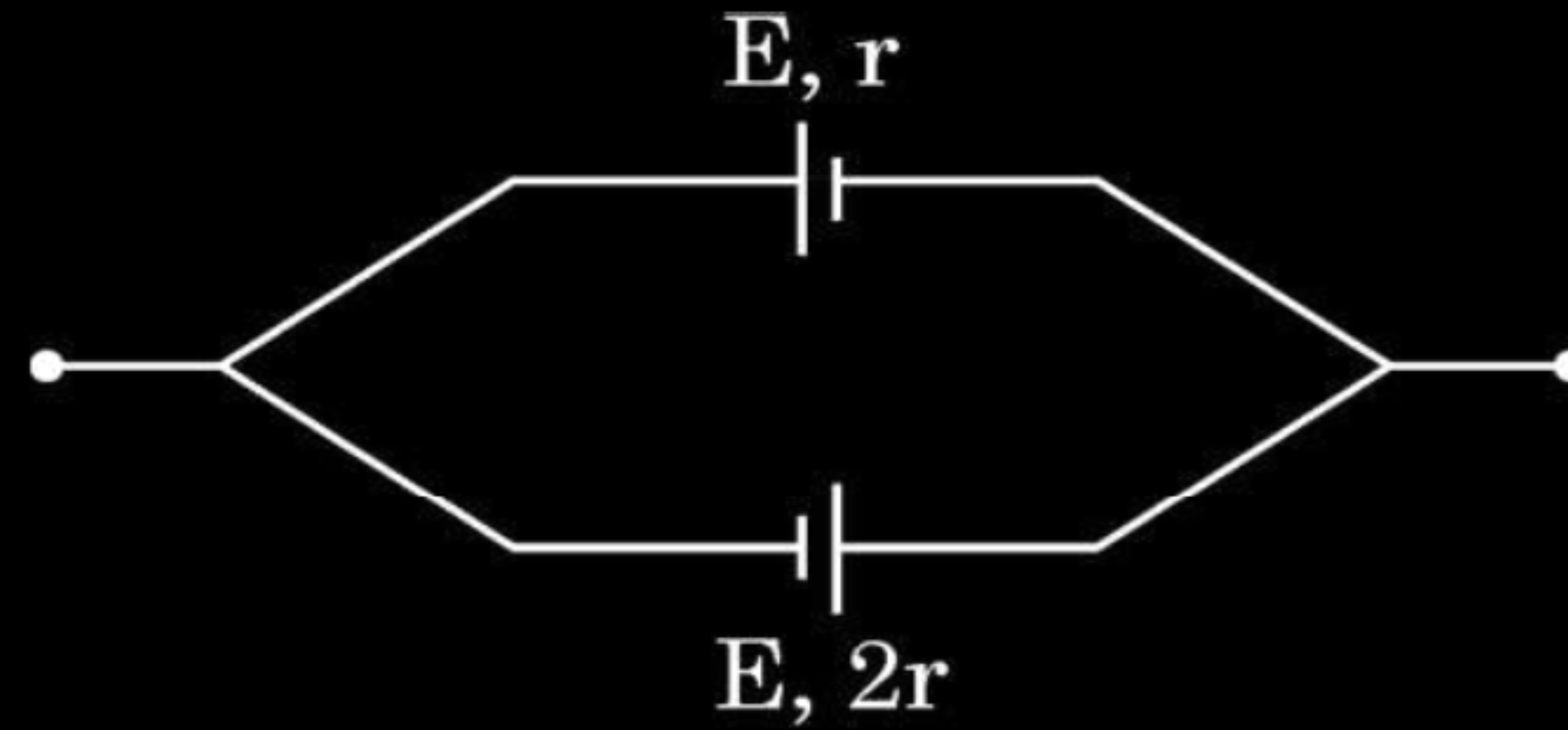
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(a) Current in the circuit.

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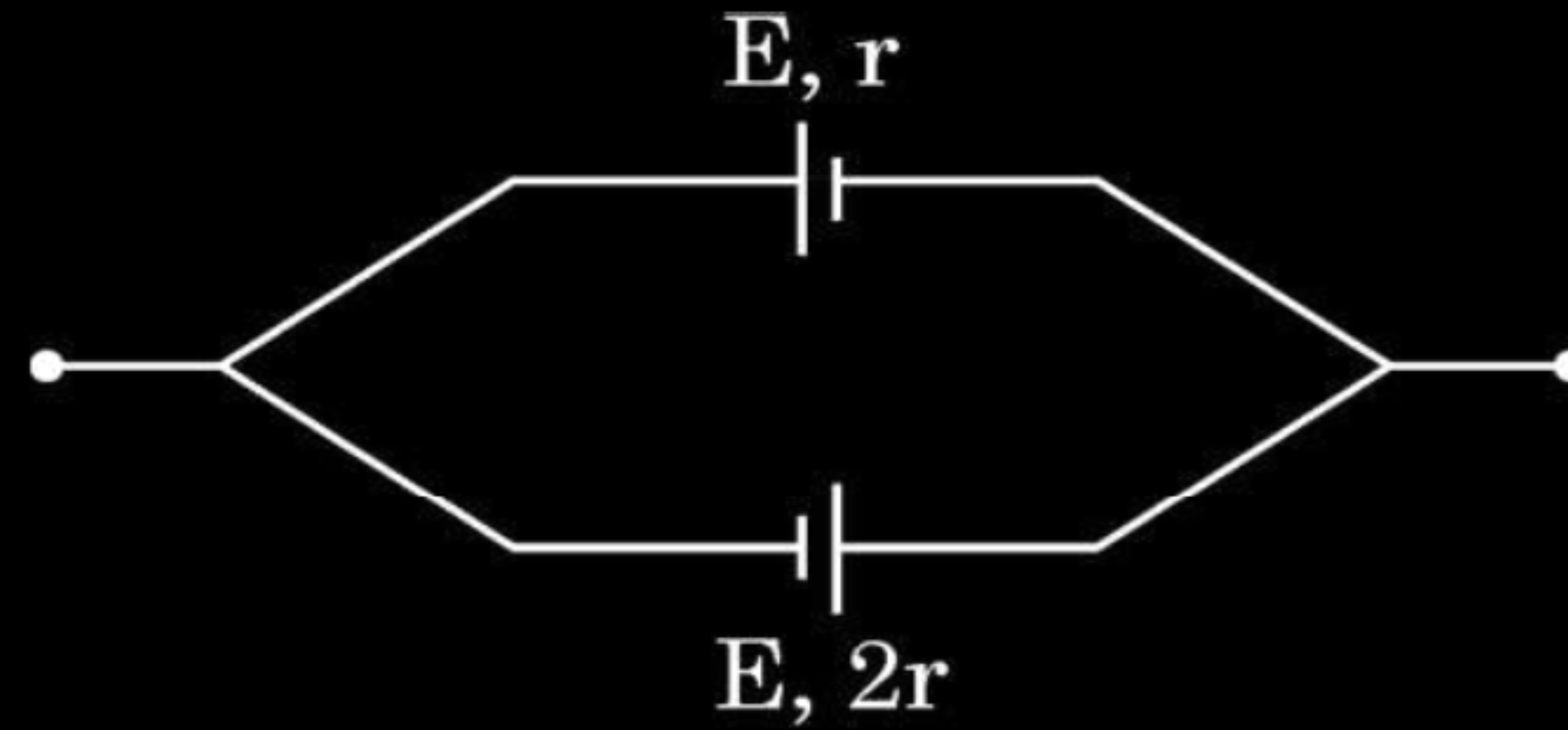
14. Two cells of emf  $E$  each and internal resistances  $r$  and  $2r$  are connected in parallel as shown in the figure. The equivalent emf of the combination is :



- |     |               |     |               |
|-----|---------------|-----|---------------|
| (a) | zero          | (b) | $\frac{E}{2}$ |
| (c) | $\frac{E}{3}$ | (d) | $E$           |



14. Two cells of emf  $E$  each and internal resistances  $r$  and  $2r$  are connected in parallel as shown in the figure. The equivalent emf of the combination is :



- |     |               |     |               |
|-----|---------------|-----|---------------|
| (a) | zero          | (b) | $\frac{E}{2}$ |
| (c) | $\frac{E}{3}$ | (d) | $E$           |

---

(c)  $\frac{E}{3}$

---



**17.**    *Assertion (A) :* The internal resistance of a cell is constant.

*Reason (R) :*    Ionic concentration of the electrolyte remains same during use of a cell.



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*Reason (R) :*    Ionic concentration of the electrolyte remains same during use of a cell.

---

(d) Assertion (A) is false and Reason (R) is also false

---



**13.** The emf and internal resistance of a cell are  $E$  and  $r$  respectively. It is connected across an external resistance  $R = 2r$ . The potential drop across the terminals of the cell will be :

(a)  $\frac{E}{4}$

(b)  $\frac{E}{2}$

(c)  $\frac{2}{3}E$

(d)  $\frac{E}{3}$



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---

(c)  $\frac{2}{3}E$

---



13. A cell of emf  $E$  is connected across an external resistance  $R$ . When current 'I' is drawn from the cell, the potential difference across the electrodes of the cell drops to  $V$ . The internal resistance 'r' of the cell is

1

(a)  $\left(\frac{E - V}{E}\right) R$

(b)  $\left(\frac{E - V}{R}\right)$

(c)  $\frac{(E - V) R}{I}$

(d)  $\left(\frac{E - V}{V}\right) R$



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(b)  $\left(\frac{E - V}{R}\right)$

(c)  $\frac{(E - V) R}{I}$

(d)  $\left(\frac{E - V}{V}\right) R$

---

( d )  $\left(\frac{E - V}{V}\right) R$

---



29. Two cells of emf  $E_1$  and  $E_2$  and internal resistances  $r_1$  and  $r_2$  are connected in parallel, with their terminals of the same polarity connected together. Obtain an expression for the equivalent emf of the combination. **3**





1/2

$$V \equiv V(B_1) - V(B_2) = \varepsilon_1 - I_1 r_1$$

1/2

$$V \equiv V(B_1) - V(B_2) = \varepsilon_2 - I_2 r_2$$

1/2

$$I = I_1 + I_2$$

1/2

$$I = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left( \frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} \right) - V \left( \frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \frac{r_1 r_2}{r_1 + r_2}$$

1/2

$$\therefore V = \varepsilon_{eq} - I r_{eq}$$

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

1/2



OR

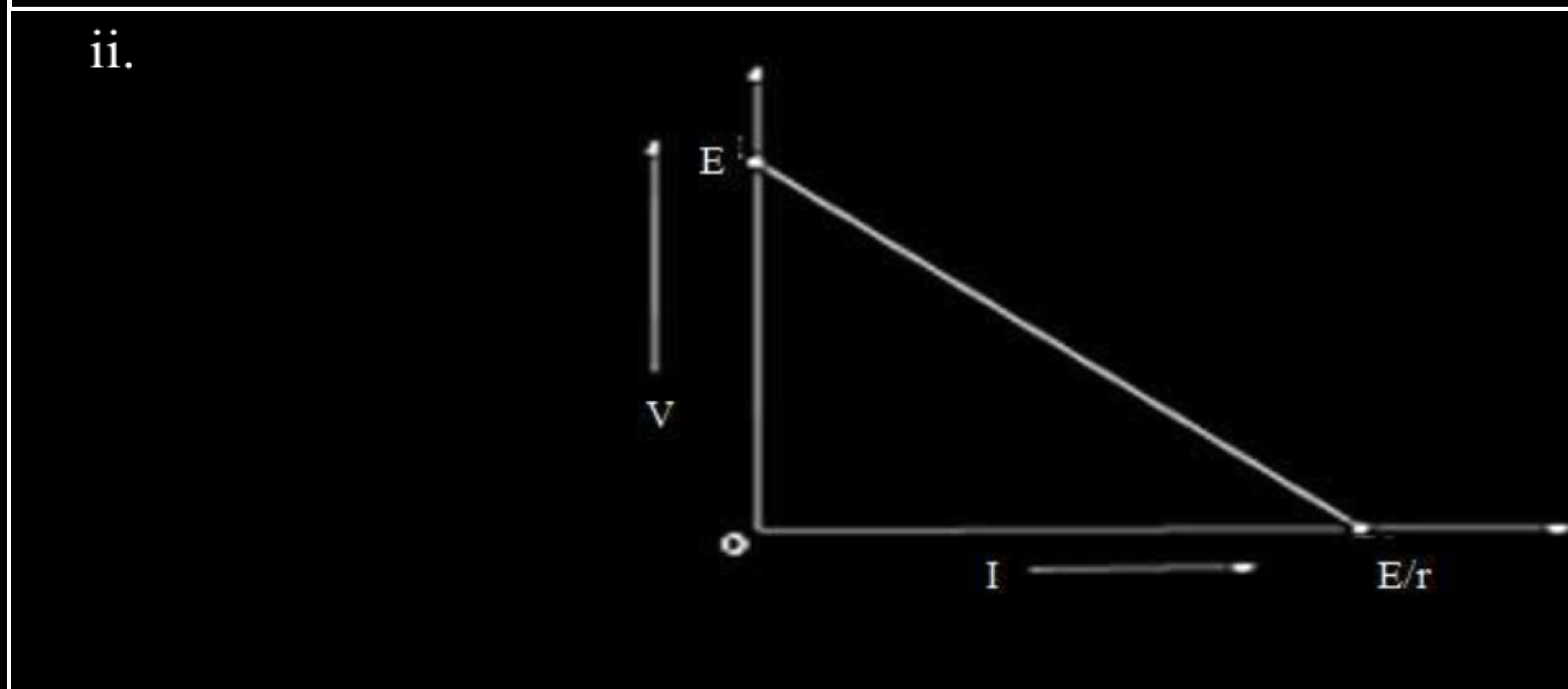
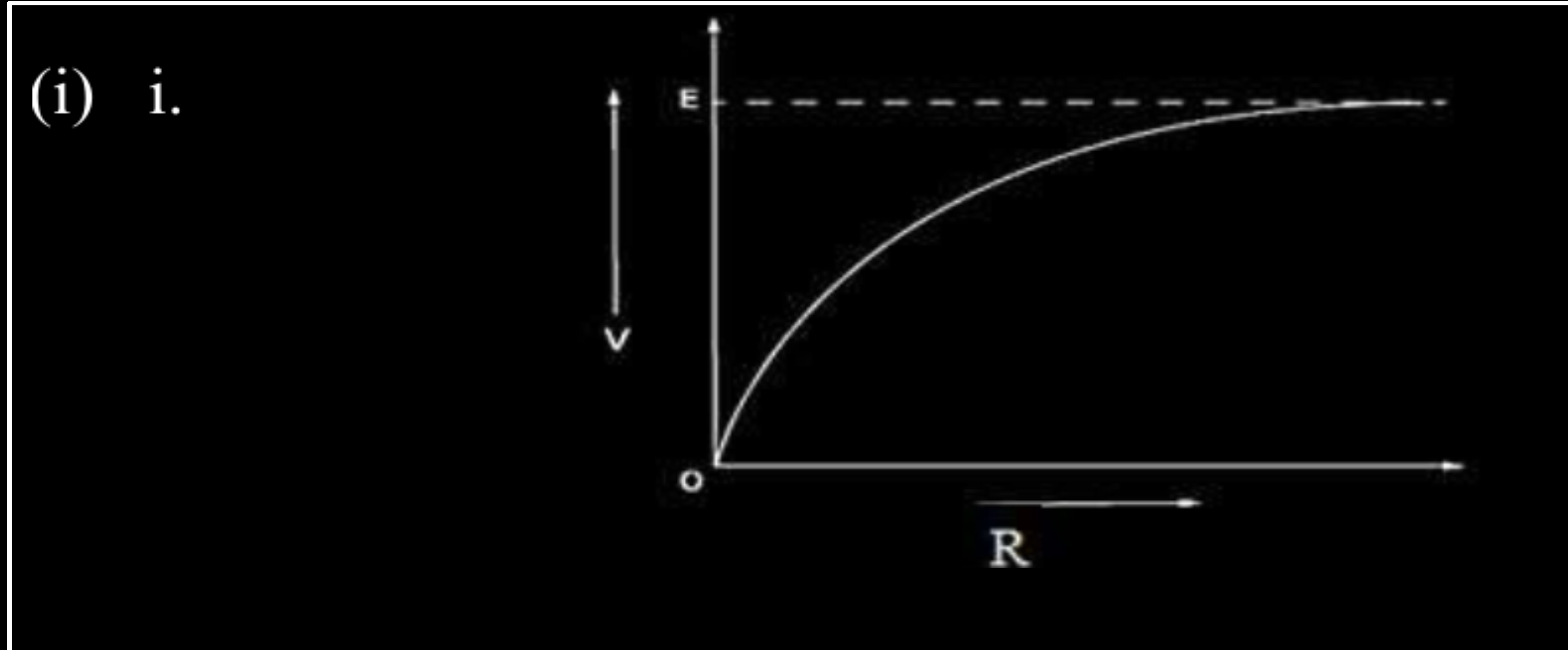
(b) (i) A cell emf of ( $E$ ) and internal resistance ( $r$ ) is connected across a variable load resistance ( $R$ ). Draw plots showing the variation of terminal voltage  $V$  with (i)  $R$  and (ii) the current ( $I$ ) in the load. 2

(ii) Three cells, each of emf  $E$  but internal resistances  $2r$ ,  $3r$  and  $6r$  are connected in parallel across a resistor  $R$ .

Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell. 3

- |                                                                                         |                  |
|-----------------------------------------------------------------------------------------|------------------|
| (i)                                                                                     |                  |
| i Plot showing variation of $V$ with $R$                                                | (1)              |
| ii Plot showing variation of $V$ with $I$                                               | (1)              |
| (ii)                                                                                    |                  |
| (i) Obtaining expression for current flowing through circuit                            | $(1\frac{1}{2})$ |
| (ii) Obtaining expression for terminal potential difference across the equivalent cell. | $(1\frac{1}{2})$ |





|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| (ii) | $\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$ $\frac{1}{r_{eq}} = \frac{1}{2r} + \frac{1}{3r} + \frac{1}{6r}$ $r_{eq} = r$ <p>Given cells are of equal emf (E) and connected in parallel, so</p> $\frac{E_{eq}}{r_{eq}} = \frac{E}{2r} + \frac{E}{3r} + \frac{E}{6r}$ $E_{eq} = E$ <p>Current flowing <math>I = \frac{E_{eq}}{R + r_{eq}}</math></p> $I = \frac{E}{R + r}$ <p>The terminal potential difference</p> $V = E_{eq} - Ir_{eq}$ $V = E - \frac{E}{(R + r)} \times r$ $V = \frac{ER}{R + r}$ |     |
|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1/2 |
|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1/2 |
|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1/2 |
|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1/2 |
|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1/2 |



4. Two cells, one of emf  $2E$  and internal resistance  $r$  and the other of emf  $E$  and internal resistance  $r/2$  are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :

(a)  $3E$

(b)  $\frac{4E}{3}$

(c)  $2E$

(d)  $\frac{3E}{4}$



4. Two cells, one of emf  $2E$  and internal resistance  $r$  and the other of emf  $E$  and internal resistance  $r/2$  are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :

(a)  $3E$

(b)  $\frac{4E}{3}$

(c)  $2E$

(d)  $\frac{3E}{4}$

---

(b)  $\frac{4E}{3}$

---



- 32.** (a) (i) Derive the relation between the current and the drift velocity of free electrons in a conductor. Briefly explain the variation of resistance of a conductor with rise in temperature.
- (ii) An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1.5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the two cases be  $\left(\frac{1}{2}\right)$  A and  $\left(\frac{1}{3}\right)$  A respectively, calculate the internal resistance of each battery.

5

**OR**

|                                                      |   |
|------------------------------------------------------|---|
| (a) (i) Derivation of relation between I and $V_d$   | 2 |
| Explanation                                          | 1 |
| (ii) Calculating internal resistance of each battery | 2 |



Total charge transported across the area A in time  $\Delta t$  is

$$\Delta Q = -neAV_d \Delta t \quad \text{-----(1)}$$

Also the amount of charge crossing area 'A' in time  $\Delta t$  is

$$\Delta Q = I \Delta t \quad \text{-----(2)}$$

Comparing equation (1) and (2)

$$I = neAV_d$$

With increase in temperature, average speed of electrons increases resulting in more frequent collisions

Hence relaxation time  $\tau$  decreases

$$\text{As } R = \frac{ml}{ne^2 \tau A}$$

Resistance increases.

$$\text{(ii) For series } I = \frac{E}{R + r}$$

$$\frac{1}{2} = \frac{3}{R + 2r}$$

$$R + 2r = 6 \quad \text{-----(1)}$$

$$\text{For parallel } \frac{1}{3} = \frac{1.5}{R + \frac{r}{2}}$$

$$2R + r = 9 \quad \text{-----(2)}$$

After solving  $r = 1 \Omega$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

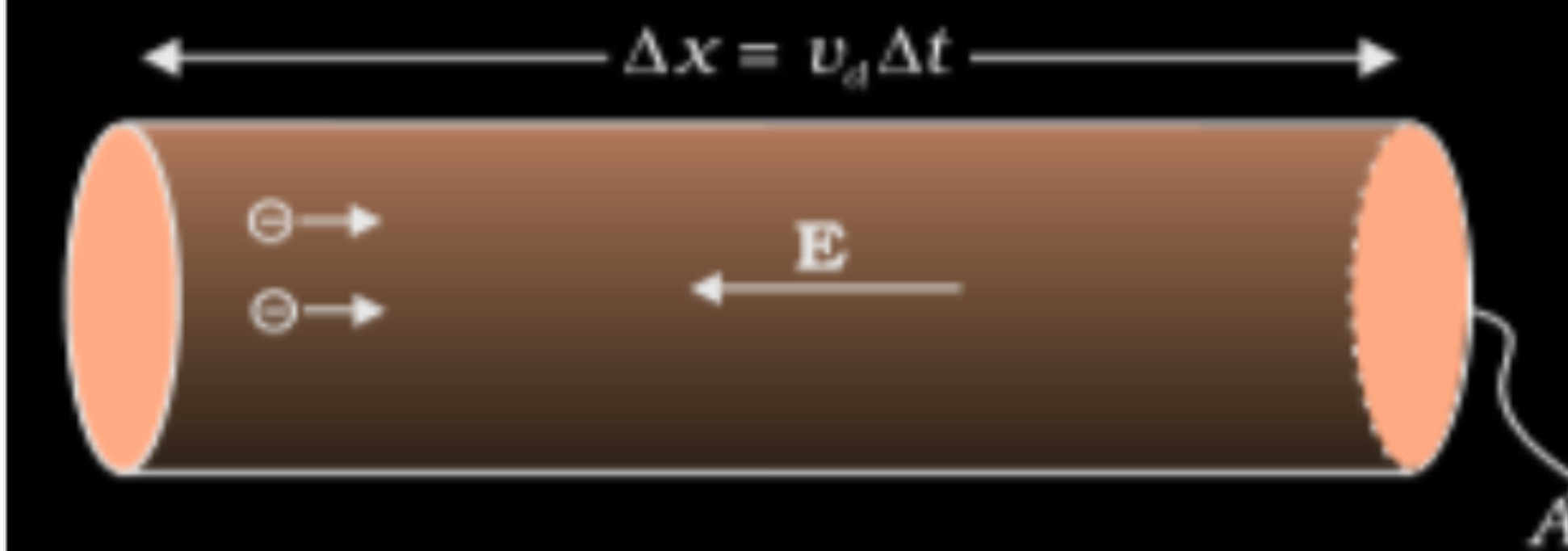
$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$





1. A battery supplies 0.9 A current through a  $2\ \Omega$  resistor and 0.3 A current through a  $7\ \Omega$  resistor when connected one by one. The internal resistance of the battery is :

(A)  $2\ \Omega$

(B)  $1.2\ \Omega$

(C)  $1\ \Omega$

(D)  $0.5\ \Omega$



1. A battery supplies 0.9 A current through a  $2\ \Omega$  resistor and 0.3 A current through a  $7\ \Omega$  resistor when connected one by one. The internal resistance of the battery is :

(A)  $2\ \Omega$

(B)  $1.2\ \Omega$

(C)  $1\ \Omega$

(D)  $0.5\ \Omega$

---

(D)  $0.5\ \Omega$

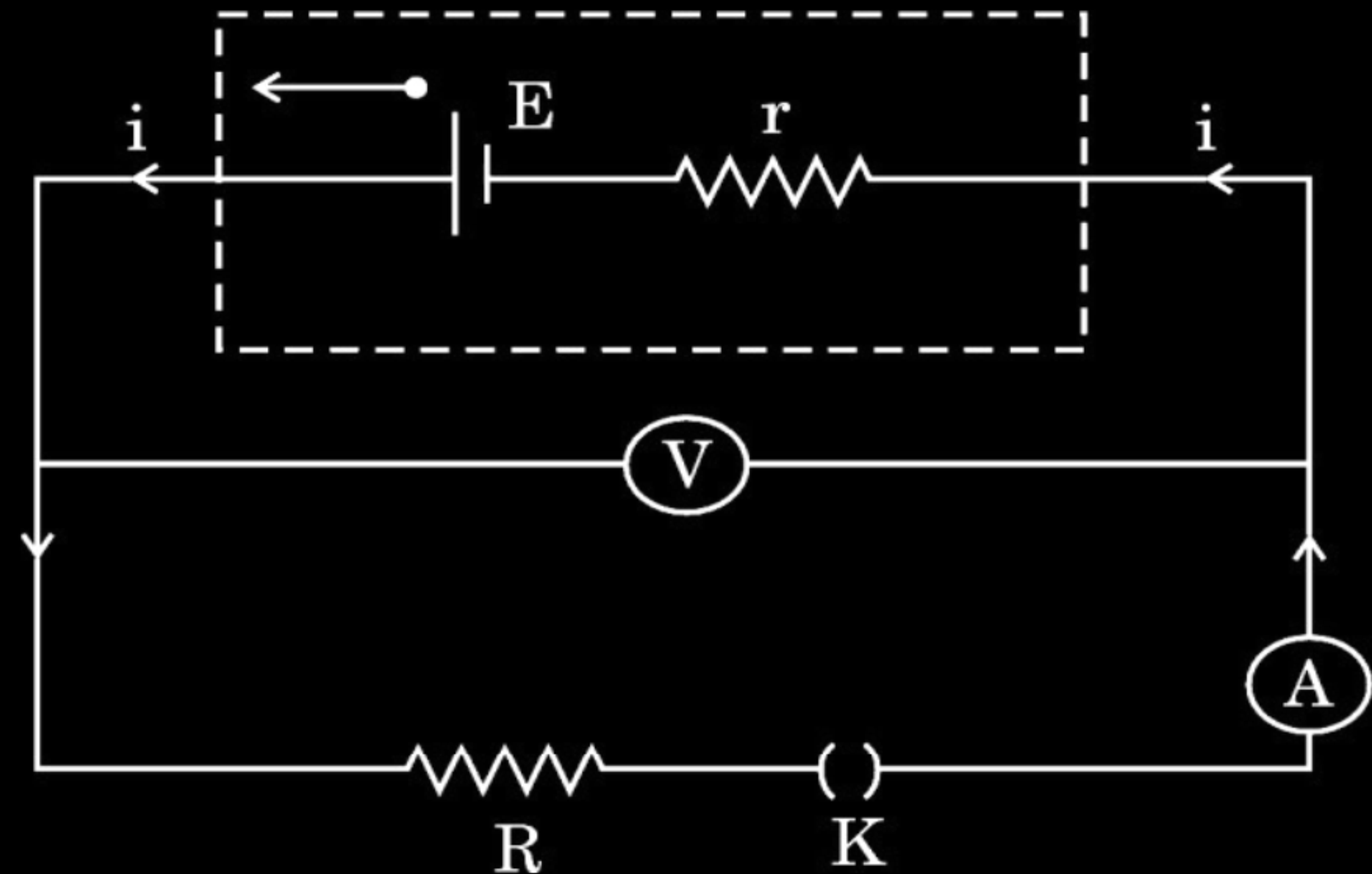
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22. A battery of unknown emf  $E$  and internal resistance  $r$  is connected in a circuit as shown in the figure. When the key ( $K$ ) is open, the voltmeter reads  $10.0\text{ V}$  and ammeter reads  $0\text{ A}$ . In the closed circuit, the voltmeter reads  $6.0\text{ V}$  and ammeter reads  $2.0\text{ A}$ . Calculate :

3

- (a) emf of the battery,
- (b) internal resistance of the battery ( $r$ ), and
- (c) external resistance ( $R$ ).



Calculation of

(a) emf of battery

$\frac{1}{2}$

(b) Internal resistance of battery( $r$ )

$1\frac{1}{2}$

(c) external resistance ( $R$ )

1



|                                                                        |               |
|------------------------------------------------------------------------|---------------|
| (a) $V = E = 10 \text{ V}$ (When key K is open and $I = 0 \text{ A}$ ) | $\frac{1}{2}$ |
| (b) $V = E - Ir$ (When key K is closed and $I = 2 \text{ A}$ )         | $\frac{1}{2}$ |
| $6 = 10 - 2r$                                                          | $\frac{1}{2}$ |
| $r = 2 \Omega$                                                         | $\frac{1}{2}$ |
| (c) $E = I(r + R)$                                                     | $\frac{1}{2}$ |
| $10 = 2(2 + R)$                                                        |               |
| $R = 3 \Omega$                                                         | $\frac{1}{2}$ |



**30.** When the terminals of a cell are connected to a conductor of resistance  $R$ , an electric current flows through the circuit. The electrolyte of the cell also offers some resistance in the path of the current, like the conductor. This resistance offered by the electrolyte is called internal resistance of the cell ( $r$ ). It depends upon the nature of the electrolyte, the area of the electrodes immersed in the electrolyte and the temperature. Due to internal resistance, a part of the energy supplied by the cell is wasted in the form of heat.

When no current is drawn from the cell, the potential difference between the two electrodes is known as emf of the cell ( $\epsilon$ ). With a current drawn from the cell, the potential difference between the two electrodes is termed as terminal potential difference ( $V$ ).



(i) Choose the *incorrect* statement : 1

- (A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf ( $\varepsilon$ ), during discharge of the cell.
- (B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.
- (C) When current is drawn from the cell then  $V = \varepsilon - Ir$ .
- (D) The graph between potential difference between the two terminals of the cell (V) and the current (I) through it is a straight line with a negative slope.

(ii) Two cells of emfs 2.0 V and 6.0 V and internal resistances 0.1  $\Omega$  and 0.4  $\Omega$  respectively, are connected in parallel. The equivalent emf of the combination will be : 1

- |           |           |
|-----------|-----------|
| (A) 2.0 V | (B) 2.8 V |
| (C) 6.0 V | (D) 8.0 V |



(i) Choose the *incorrect* statement : 1

(A) The potential difference ( $V$ ) between the two terminals of a cell in a closed circuit is always less than its emf ( $\varepsilon$ ), during discharge of the cell.

(B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(C) When current is drawn from the cell then  $V = \varepsilon - Ir$ .

(D) The graph between potential difference between the two terminals of the cell ( $V$ ) and the current ( $I$ ) through it is a straight line with a negative slope.

(ii) Two cells of emfs  $2.0\text{ V}$  and  $6.0\text{ V}$  and internal resistances  $0.1\ \Omega$  and  $0.4\ \Omega$  respectively, are connected in parallel. The equivalent emf of the combination will be : 1

(A)  $2.0\text{ V}$

(B)  $2.8\text{ V}$

(C)  $6.0\text{ V}$

(D)  $8.0\text{ V}$



- (iii) Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential  $V_+$  ( $V_+ > 0$ ), and the negative electrode develops a potential  $- (V_-)$  ( $V_- \geq 0$ ), relative to the electrolyte adjacent to it. When no current is drawn from the cell then : 1

- |                                   |                                   |
|-----------------------------------|-----------------------------------|
| (A) $\varepsilon = V_+ + V_- > 0$ | (B) $\varepsilon = V_+ - V_- > 0$ |
| (C) $\varepsilon = V_+ + V_- < 0$ | (D) $\varepsilon = V_+ + V_- = 0$ |

- (iv) (a) Five identical cells, each of emf 2 V and internal resistance  $0.1 \Omega$  are connected in parallel. This combination in turn is connected to an external resistor of  $9.98 \Omega$ . The current flowing through the resistor is : 1

- |            |           |
|------------|-----------|
| (A) 0.05 A | (B) 0.1 A |
| (C) 0.15 A | (D) 0.2 A |

**OR**

- (b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is : 1

- |                  |                  |
|------------------|------------------|
| (A) $1.0 \Omega$ | (B) $1.5 \Omega$ |
| (C) $2.0 \Omega$ | (D) $2.5 \Omega$ |



- (iii) Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential  $V_+$  ( $V_+ > 0$ ), and the negative electrode develops a potential  $- (V_-)$  ( $V_- \geq 0$ ), relative to the electrolyte adjacent to it. When no current is drawn from the cell then : 1

(A)  $\varepsilon = V_+ + V_- > 0$

(B)  $\varepsilon = V_+ - V_- > 0$

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(A) 0.05 A

(B) 0.1 A

(C) 0.15 A

(D) 0.2 A

**OR**

- (b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is : 1

(A)  $1.0 \Omega$

(B)  $1.5 \Omega$

(C)  $2.0 \Omega$

(D)  $2.5 \Omega$



**3.** A student is asked to connect four cells, each of emf  $E$  and internal resistance  $r$ , in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be :

(A)  $4E$  and  $2r$

(B)  $4E$  and  $3r$

(C)  $3E$  and  $4r$

(D)  $2E$  and  $4r$



**3.** A student is asked to connect four cells, each of emf  $E$  and internal resistance  $r$ , in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be :

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(B)  $4E$  and  $3r$

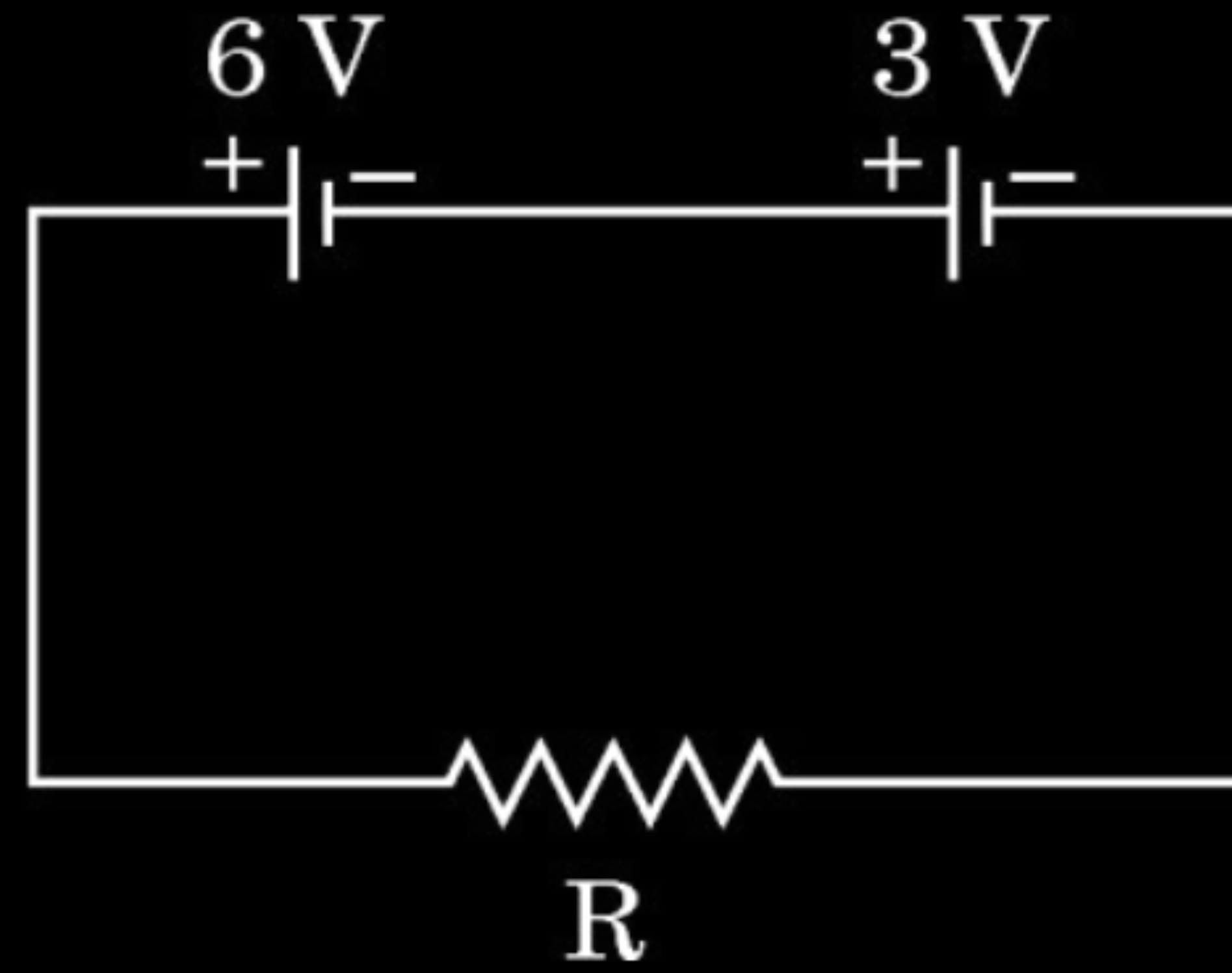
(C)  $3E$  and  $4r$

(D)  $2E$  and  $4r$

(D)  $2E$  and  $4r$



17. Two batteries of emfs 6 V and 3 V and internal resistances  $0.8\ \Omega$  and  $0.2\ \Omega$  respectively are connected in series to an external resistance  $R$ , as shown in figure. Find the value of  $R$  so that the potential difference across the 6 V battery be zero.





Finding the value of R

2

$$i = \frac{9}{R+1}$$

As potential difference across 6V is zero;

$$6 - ir = 0 \Rightarrow 6 - \left( \frac{9}{R+1} \right) (0.8) = 0$$

On solving;

$$R = 0.2\Omega$$

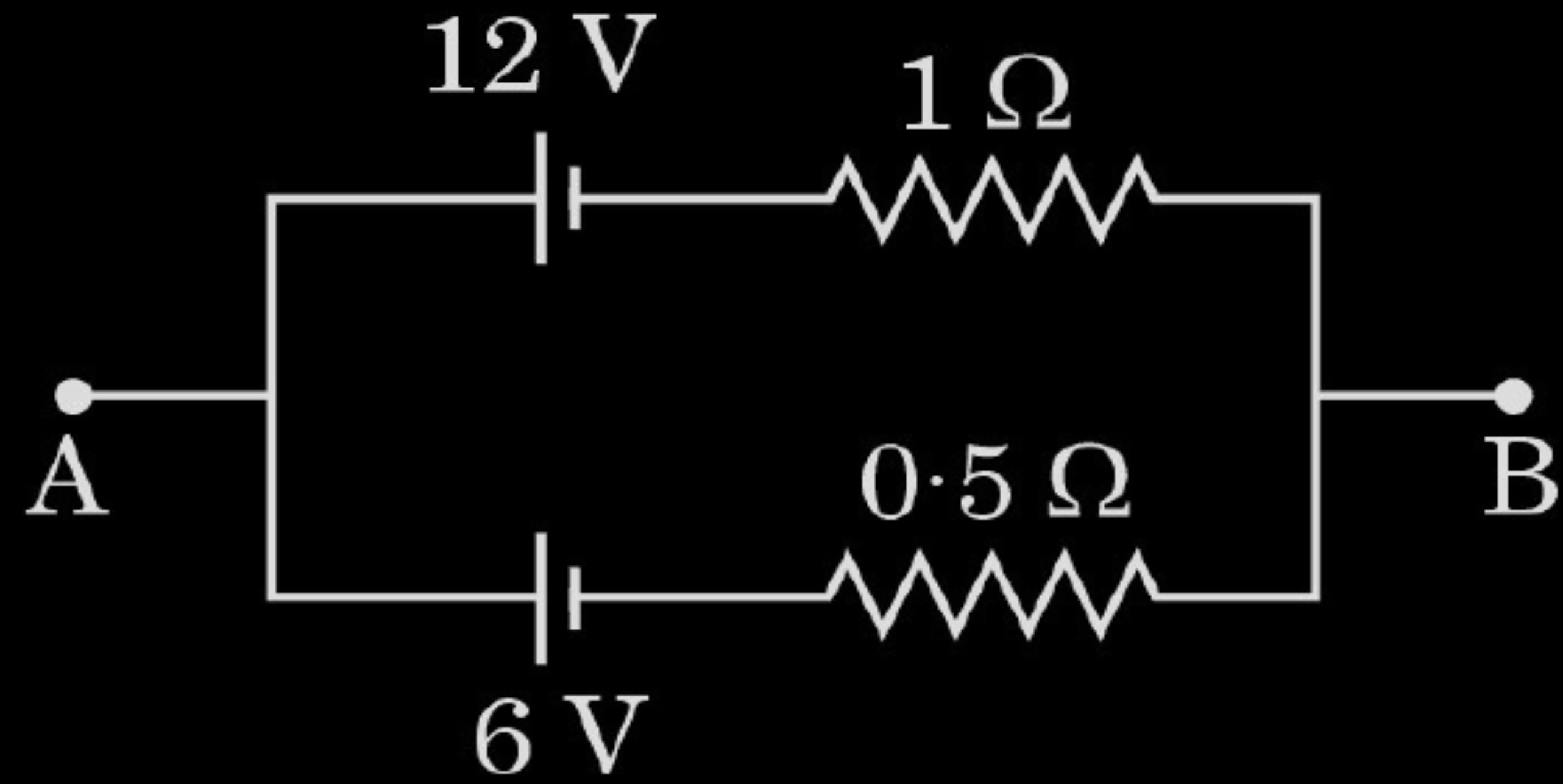
$\frac{1}{2}$

$\frac{1}{2}$

1



3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V

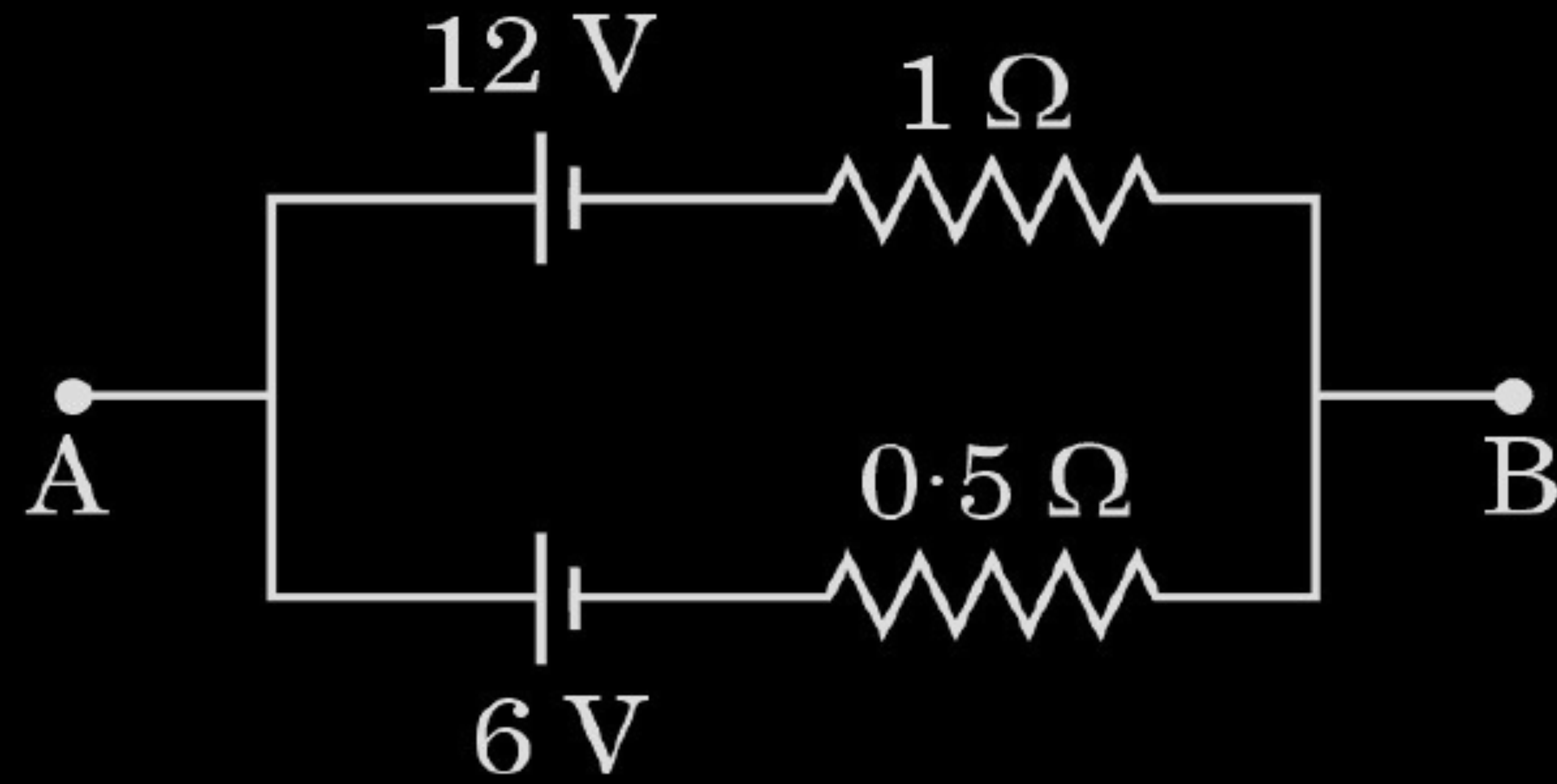
(B) 8 V

(C) 9 V

(D) 12 V



3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

(B) 8V



**2.** The emf and internal resistance of a cell are 3 V and  $0.2\ \Omega$  respectively. It is connected to an external resistor of  $5.8\ \Omega$ . The potential difference across the cell will be :

(A)  $1.2\text{ V}$

(B)  $2.0\text{ V}$

(C)  $2.9\text{ V}$

(D)  $3.2\text{ V}$



**2.** The emf and internal resistance of a cell are 3 V and  $0.2\ \Omega$  respectively. It is connected to an external resistor of  $5.8\ \Omega$ . The potential difference across the cell will be :

(A) 1.2 V

(B) 2.0 V

(C) 2.9 V

(D) 3.2 V

(C) 2.9 V



- 17.** Two identical cells, each of emf  $E$  and internal resistance  $r$ , are connected in series with a resistor of  $8\ \Omega$ . A current of  $1.5\text{ A}$  flows in the circuit. When these cells are connected in parallel, they send  $1.0\text{ A}$  current through the same resistor. Calculate the internal resistance,  $r$  and emf,  $E$  of the cell.

2

|                                        |   |
|----------------------------------------|---|
| Calculation of (i) Internal Resistance | 1 |
| (ii) EMF                               | 1 |



### In Series Combination

$$\begin{aligned} 2E &= I(2r+8) \\ &= 1.5(2r+8) \end{aligned} \dots\dots\dots(1)$$

### In Parallel Combination

$$\begin{aligned} E &= I\left(\frac{r}{2} + 8\right) \\ E &= 1\left(\frac{r}{2} + 8\right) \end{aligned} \dots\dots\dots(2)$$

Solving equation (1) & (2)

$$2 = \frac{1.5(2r+8)}{\left(\frac{r}{2} + 8\right)}$$

$$r + 16 = 3r + 12$$

$$2r = 4$$

$$r = 2\Omega$$

Substituting the value of r in equation(2)

$$E = \left(\frac{2}{2} + 8\right)$$

$$E = 9V$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$



19. (a) A cell is connected across an external resistance  $12\ \Omega$  and supplies  $0.25\ \text{A}$  current. When the external resistance is increased by  $4\ \Omega$ , the current reduces to  $0.2\ \text{A}$ . Calculate (i) the emf, and (ii) the internal resistance, of the cell.



19. (a) A cell is connected across an external resistance  $12\ \Omega$  and supplies  $0.25\ \text{A}$  current. When the external resistance is increased by  $4\ \Omega$ , the current reduces to  $0.2\ \text{A}$ . Calculate (i) the emf, and (ii) the internal resistance, of the cell.

2

|                                              |   |
|----------------------------------------------|---|
| (i) Calculating e.m.f of cell                | 1 |
| (ii) Calculating internal resistance of cell | 1 |

$$I = \frac{\mathcal{E}}{R + r}$$

$$0.25 = \frac{\mathcal{E}}{12 + r} \quad \text{----- (1)}$$

$$0.2 = \frac{\mathcal{E}}{16 + r} \quad \text{----- (2)}$$

On solving eq (1) and (2)

$$r = 4\ \Omega$$

$$\mathcal{E} = 4\ \text{V}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



22. (a) Define mobility of free electrons in a conductor. How does the mobility of electrons in a metal change when :

(i) temperature of the conductor is decreased at constant potential difference ?

(ii) applied potential difference is doubled at a constant temperature ?

3

**OR**

(b) Obtain an expression for emf and internal resistance of a cell which is equivalent to parallel combination of two cells of emf and internal resistance ( $E_1, r_1$ ) and ( $E_2, r_2$ ), respectively.

3

|                                           |   |
|-------------------------------------------|---|
| • Definition of mobility                  | 1 |
| • Change in mobility when                 |   |
| (i) Temperature of conductor is decreased | 1 |
| (ii) Applied voltage is doubled           | 1 |

|                                         |   |
|-----------------------------------------|---|
| Obtaining expression for equivalent emf | 2 |
| Equivalent resistance                   | 1 |



Mobility of free electrons in a conductor is defined as the magnitude of the drift velocity per unit electric field.

1

Alternatively:-

$$\mu = \frac{|\vec{v}_d|}{E}$$

(i) When the temperature of the conductor is decreased at constant potential difference the mobility will increase.

1

(ii) When the applied potential difference is doubled at a constant temperature the mobility will be unchanged.

1

Alternatively:-

(i) 
$$\mu = \frac{e\tau}{m}$$

As temperature decreases  $\tau$  will increase and  $\mu$  will also increase.

(ii) 
$$\mu = \frac{e\tau}{m}$$

As  $\mu$  is independent of applied voltage hence  $\mu$  remains unchanged.



When two cells are connected in parallel the total current

$$I = I_1 + I_2 \text{ -----(1)}$$

$$I = \frac{\mathcal{E}_1 - V}{r_1} + \frac{\mathcal{E}_2 - V}{r_2}$$

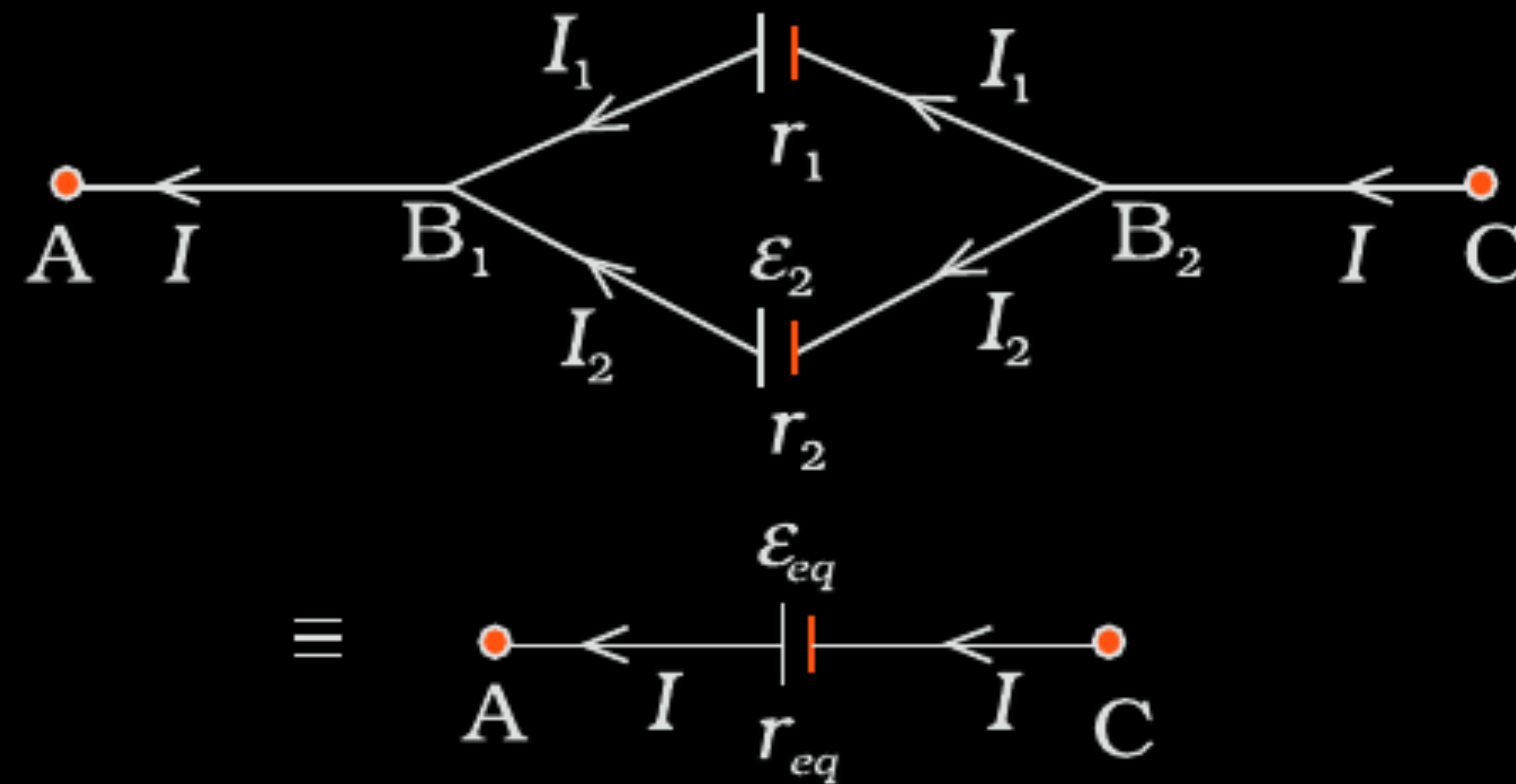
$$I = \left( \frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2} \right) - V \left( \frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$V = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2} - I \left( \frac{r_1 r_2}{r_1 + r_2} \right)$$

Hence

$$\mathcal{E}_{eq} = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$



1/2

1/2

1/2

1/2

1/2

1/2



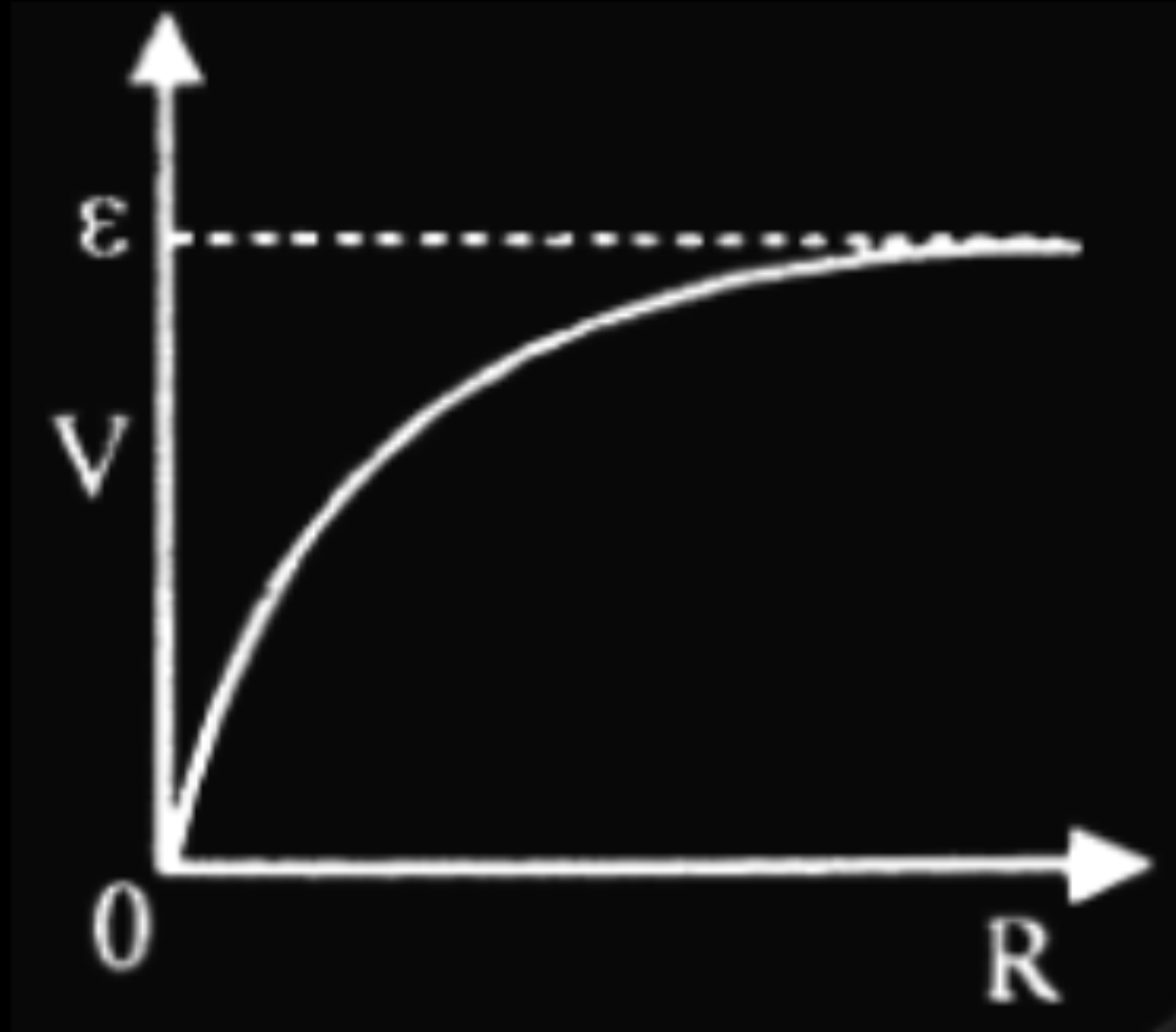
- 17.** A cell of emf  $E$  and internal resistance  $r$  is connected across a resistor of variable resistance  $R$ . Show graphically the variation of
- (a) the terminal voltage across the cell,
  - (b) the current supplied by the cell,
- with  $R$  as it is increased from 0 to the maximum value.



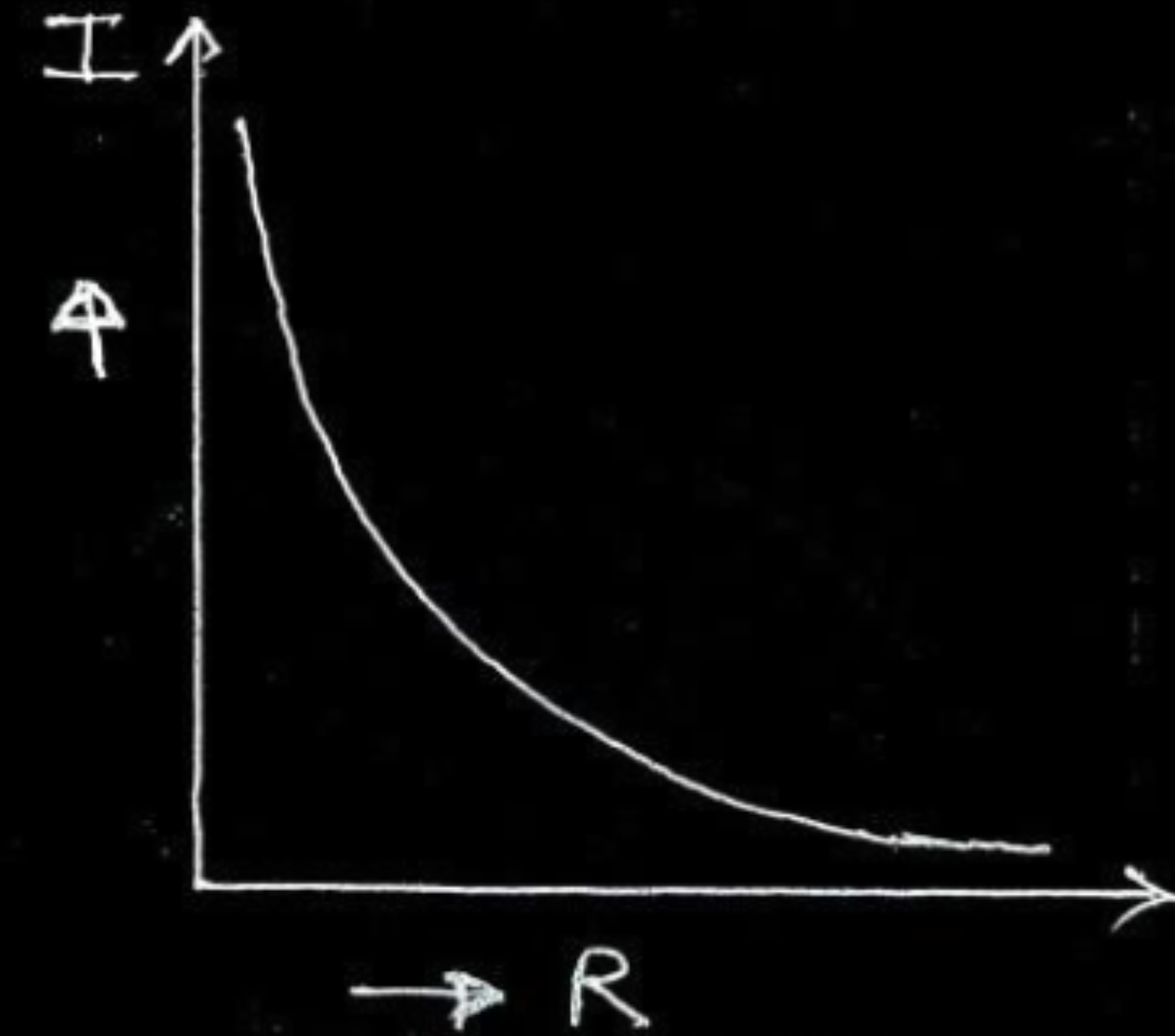
17. A cell of emf  $E$  and internal resistance  $r$  is connected across a resistor of variable resistance  $R$ . Show graphically the variation of
- (a) the terminal voltage across the cell,
  - (b) the current supplied by the cell,
- with  $R$  as it is increased from 0 to the maximum value.

2

(a)



(b)



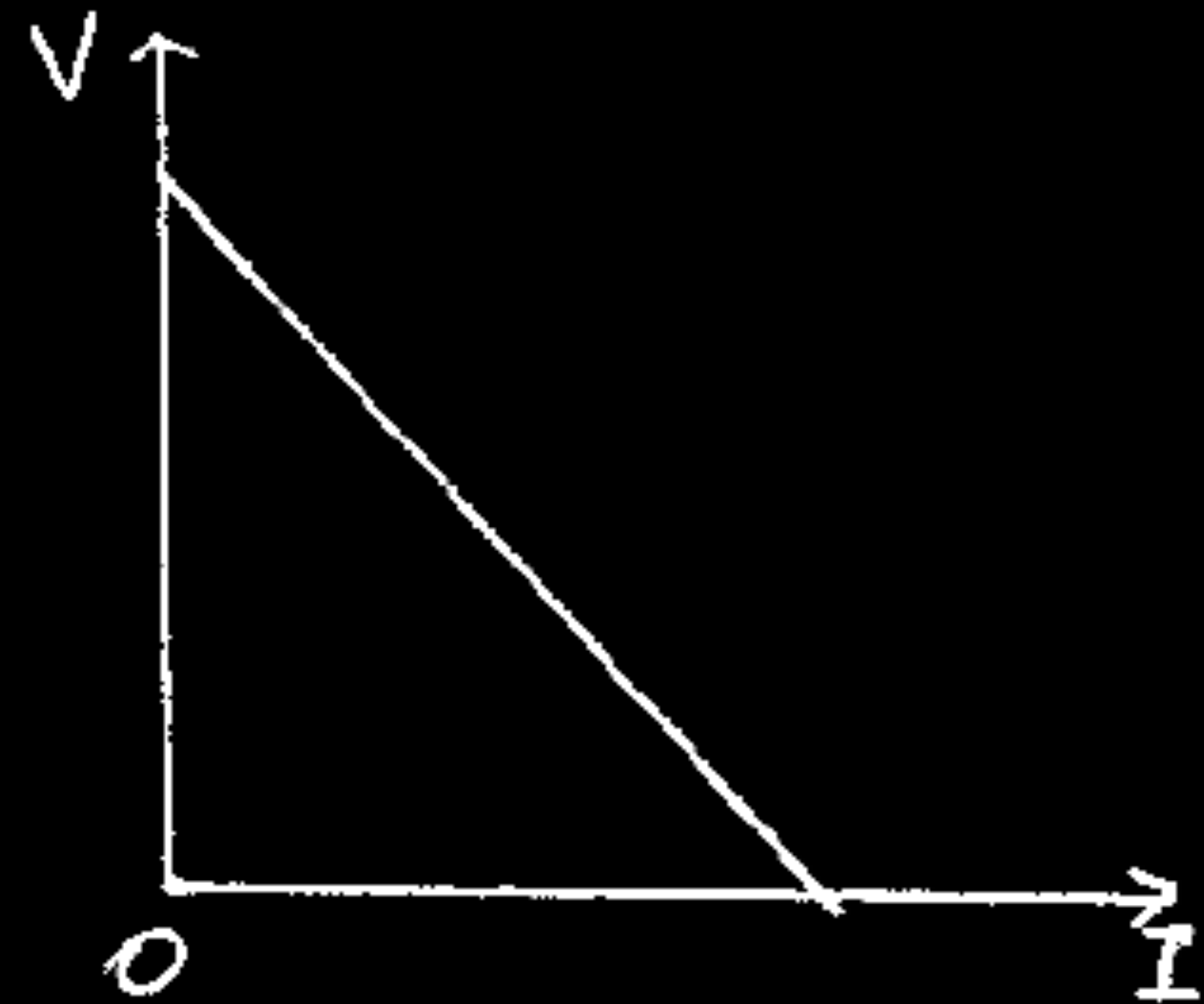


- 17.** A cell of emf  $E$  and internal resistance  $r$  is connected to an external variable resistance  $R$ . Plot a graph showing the variation of terminal voltage  $V$  of the cell as a function of current  $I$ , supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it.



17. A cell of emf  $E$  and internal resistance  $r$  is connected to an external variable resistance  $R$ . Plot a graph showing the variation of terminal voltage  $V$  of the cell as a function of current  $I$ , supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it.

2



$$V = E - Ir$$

$E$  = intercept on y-axis (i.e. V-axis)

$r$  = |slope of graph|

1

$\frac{1}{2}$

$\frac{1}{2}$



22. (a) Define the terms 'emf' and 'terminal voltage' of a cell. Can the terminal voltage of a cell ever exceed its emf? Explain. **3**

**OR**

(b) Define the term current density. Write its SI unit. Derive the equivalent form of Ohm's law  $\vec{j} = \sigma \vec{E}$ . **3**

- |                                                 |     |
|-------------------------------------------------|-----|
| • Defining emf and terminal voltage             | 1+1 |
| • Explaining terminal voltage exceeding the emf | 1   |

- |                                                  |      |
|--------------------------------------------------|------|
| Defining current density and Writing its SI unit | 1+ ½ |
| Deriving $\vec{J} = \sigma \vec{E}$              | 1 ½  |



|                                                                                                                            |               |
|----------------------------------------------------------------------------------------------------------------------------|---------------|
| <p><b>emf</b>- It is the potential difference across the terminal of the cell, when no current is being drawn from it.</p> | 1             |
| <p><b>Terminal Voltage</b> - It is the potential difference between the terminals of the cell in a closed circuit.</p>     | 1             |
| <p><b>Alternatively</b> <math>V = \varepsilon - Ir</math></p>                                                              | $\frac{1}{2}$ |
| <p>Yes,</p> <p>During charging, <math>V = \varepsilon - (-I)r = \varepsilon + Ir</math></p>                                | $\frac{1}{2}$ |
| <p><b>Current density</b> – It is defined as the current through unit area normal to the current.</p>                      | 1             |
| <p><b>Alternatively,</b> <math>\vec{J} = \frac{I}{\vec{A}}</math></p>                                                      |               |
| <p>S.I unit     <math>A/m^2</math></p>                                                                                     | $\frac{1}{2}$ |
| <p><b>Derivation</b></p>                                                                                                   |               |
| $V = IR$                                                                                                                   | $\frac{1}{2}$ |
| $\vec{E}l = I \frac{\rho l}{\vec{A}}$                                                                                      | $\frac{1}{2}$ |
| $\vec{E} = \vec{J} \rho$                                                                                                   |               |
| $\vec{J} = \frac{I}{\rho} \vec{E} = \sigma \vec{E}$                                                                        | $\frac{1}{2}$ |

|                                                  |                    |
|--------------------------------------------------|--------------------|
| <p><b>Alternatively</b></p>                      | $I = ne\vec{A}v_d$ |
| $\frac{I}{\vec{A}} = \frac{ne^2\tau\vec{E}}{m}$  |                    |
| $\vec{J} = \frac{\vec{E}}{\rho} = \sigma\vec{E}$ |                    |



OR

- (b) (i) A battery of emf  $E$  and internal resistance  $r$  is connected to a variable external resistance  $R$ .
- (I) Obtain the expression for current  $I$  in the circuit and the value of maximum current the battery can supply.
- (II) Obtain the terminal voltage  $V$  across the battery and its maximum possible value.
- (ii) The above battery sends a current  $I_1$  when  $R = R_1$  and a current  $I_2$  when  $R = R_2$ . Obtain the internal resistance of the battery in terms of  $I_1$ ,  $I_2$ ,  $R_1$  and  $R_2$ .

3

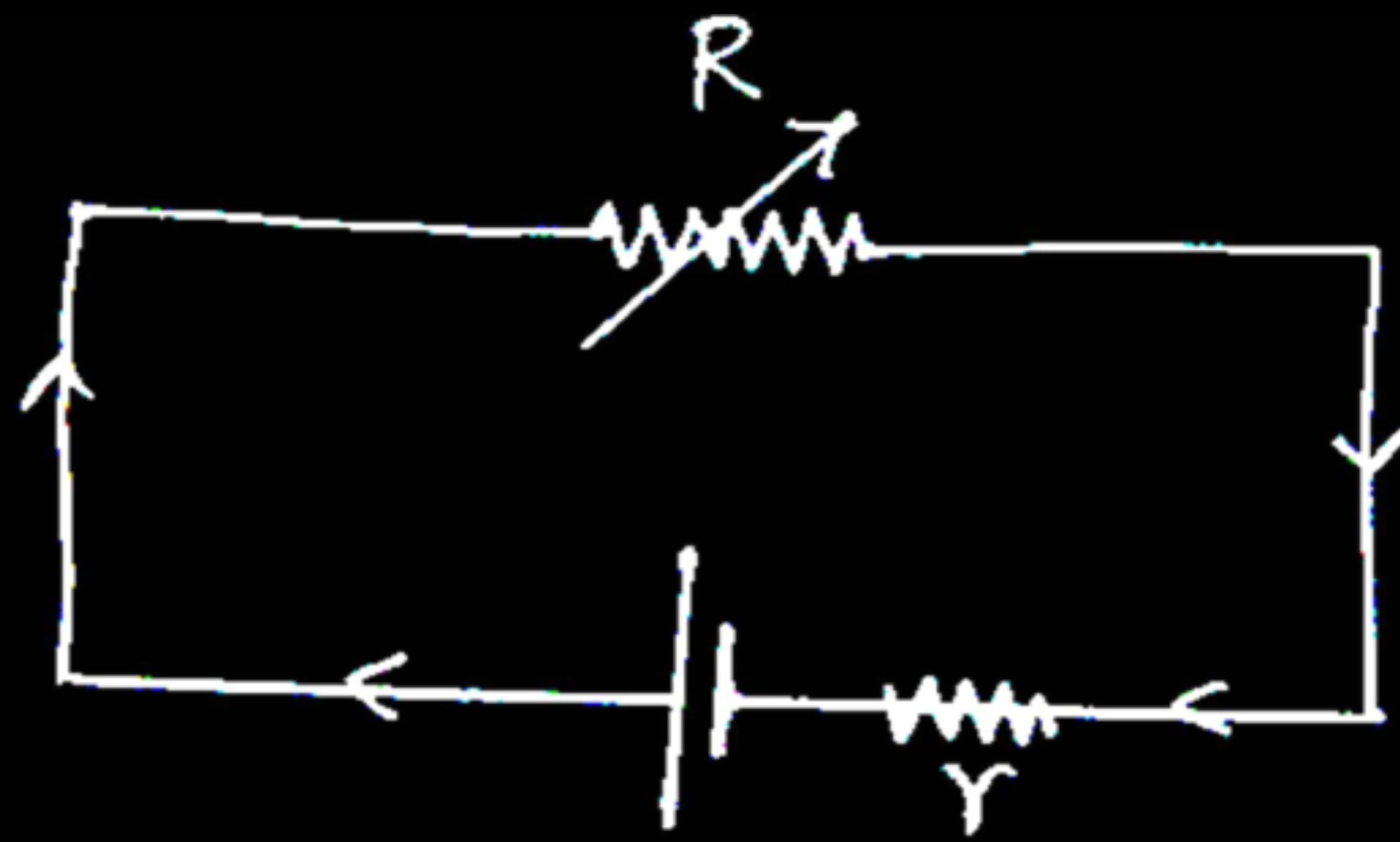
(i) (I)

- Obtaining the expression for current  $\frac{1}{2}$
- Finding value of maximum current  $\frac{1}{2}$

(II) Obtaining terminal voltage  $V$  and its maximum possible value 1

(ii) Obtaining the internal resistance of the battery 1





- $V = E - Ir$   
 $IR = E - Ir$   
 $I = \frac{E}{R + r}$

For maximum value of current  $R=0$

$$I_{\max} = \frac{E}{r}$$

(II)  $V = V_+ + V_- - Ir$   
 $V = E - Ir$

$V_{\max} = E$  , when  $I=0$

(ii)  $I_1 R_1 + I_1 r = I_2 R_2 + I_2 r$

$$r = \frac{I_2 R_2 - I_1 R_1}{I_1 - I_2}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



2. A battery of e.m.f.  $12\text{ V}$  and internal resistance  $0.5\ \Omega$  is connected to a  $9.5\ \Omega$  resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is

(A)  $1.05$

(B)  $1$

(C)  $0.95$

(D)  $1.1$



2. A battery of e.m.f.  $12\text{ V}$  and internal resistance  $0.5\ \Omega$  is connected to a  $9.5\ \Omega$  resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is

(A)  $1.05$

(C)  $0.95$

(B)  $1$

(D)  $1.1$



22. (a) Define resistivity of a conductor. Discuss its dependence on temperature of the conductor and draw a plot of resistivity of copper as a function of temperature.

3

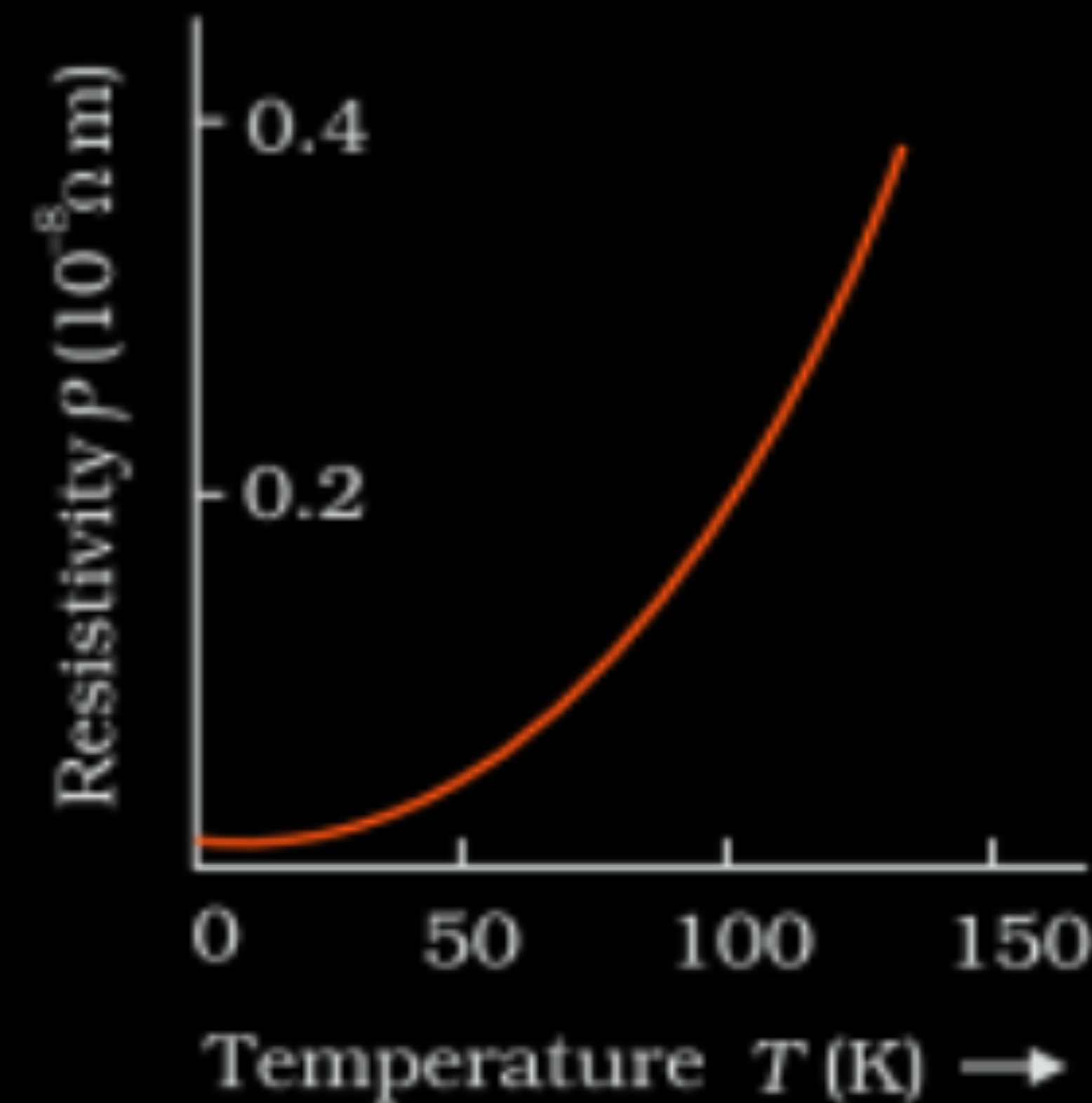
- (b) (i) “A low voltage battery from which high current is required must have low internal resistance.” Justify.
- (ii) “A high voltage battery must have a large internal resistance.” Justify.

|                                                           |               |
|-----------------------------------------------------------|---------------|
| (a) Defining resistivity                                  | 1             |
| Discussing its dependence on temperature                  | $\frac{1}{2}$ |
| Plotting graph of resistivity with temperature for copper | $\frac{1}{2}$ |
| (b) (i) Justification                                     | $\frac{1}{2}$ |
| (ii) Justification                                        | $\frac{1}{2}$ |



(a) Resistivity is the resistance of a material of unit length having unit area of cross-section.

On increasing the temperature of a conductor, the resistivity increases.



**Note:** Full credit to be given if values are not shown on the graph.

(b) (i) A low internal resistance allows large current to be drawn even at a low voltage.

(ii) To limit the current.



22. (a) A cell of e.m.f.  $E$  and internal resistance  $r$  is connected with a variable external resistance  $R$  and a voltmeter showing potential drop  $V$  across  $R$ . Obtain the relationship between  $V$ ,  $E$ ,  $R$  and  $r$ .
- (b) Draw the shape of the graph showing the variation of terminal voltage  $V$  of the cell as a function of current  $I$  drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?



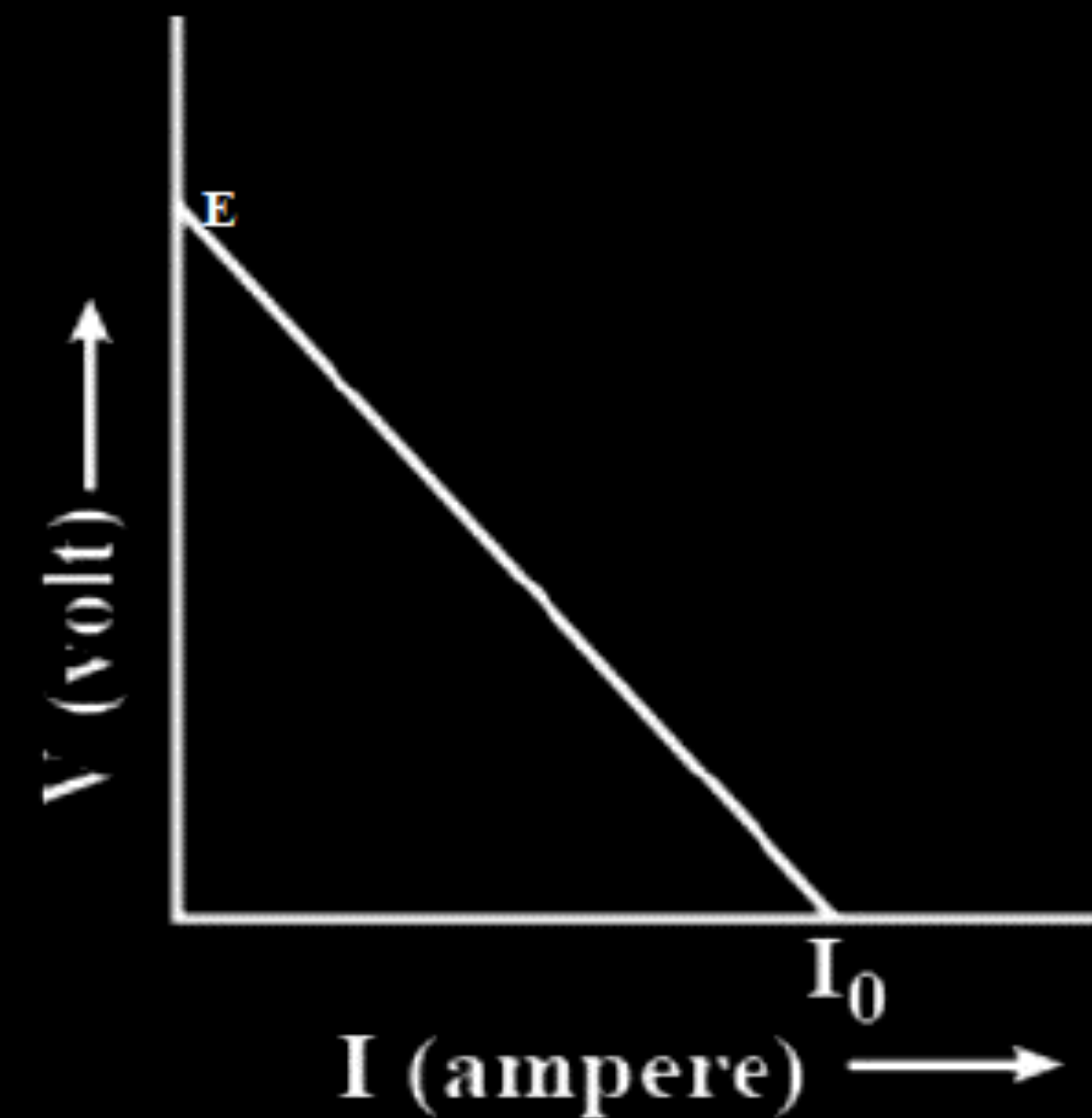
22. (a) A cell of e.m.f.  $E$  and internal resistance  $r$  is connected with a variable external resistance  $R$  and a voltmeter showing potential drop  $V$  across  $R$ . Obtain the relationship between  $V$ ,  $E$ ,  $R$  and  $r$ . 3
- (b) Draw the shape of the graph showing the variation of terminal voltage  $V$  of the cell as a function of current  $I$  drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?

(a) Current drawn from the cell  $(I) = \frac{E}{r+R}$

Potential difference  $(V) = IR$

$$V = \frac{ER}{r+R}$$

(b)



Slope of the graph is equal to internal resistance of the cell.

The intercept on Y-axis gives the emf of the cell.



17.  $n$  identical cells, each of e.m.f.  $E$  and internal resistance  $r$ , are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. 2



17.  $n$  identical cells, each of e.m.f.  $E$  and internal resistance  $r$ , are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. 2

|                                                                                                                                                                   |                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Calculating the potential difference                                                                                                                              | 2                                                               |
| <p>Net e.m.f = <math>(n-4)E</math></p> <p><math>\therefore I = \frac{(n-4)E}{nr}</math></p> <p>Potential difference across 'X'</p> <p><math>V = E + Ir</math></p> | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> |



22. (a) Two batteries of emf's 3V & 6V and internal resistances  $0.2\ \Omega$  &  $0.4\ \Omega$  are connected in parallel. This combination is connected to a  $4\ \Omega$  resistor. Find :

**3**

- (i) the equivalent emf of the combination
- (ii) the equivalent internal resistance of the combination
- (iii) the current drawn from the combination

**OR**



22. (a) Two batteries of emf's 3V & 6V and internal resistances  $0.2\ \Omega$  &  $0.4\ \Omega$  are connected in parallel. This combination is connected to a  $4\ \Omega$  resistor. Find :

3

- (i) the equivalent emf of the combination  
(ii) the equivalent internal resistance of the combination  
(iii) the current drawn from the combination

|                                                                 |               |
|-----------------------------------------------------------------|---------------|
| (i) Because $E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$      | $\frac{1}{2}$ |
| $E_{eq} = \frac{3 \times 0.4 + 6 \times 0.2}{0.6} = 4\text{ V}$ | $\frac{1}{2}$ |
| (ii) $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$                       | $\frac{1}{2}$ |
| $r_{eq} = \frac{0.2 \times 0.4}{0.2 + 0.4} = 0.133\Omega$       | $\frac{1}{2}$ |
| (iii) $I = \frac{E}{R + r_{eq}}$                                | $\frac{1}{2}$ |
| $I = \frac{4}{4 + 0.13} = \frac{4}{4.13}\text{ A}$              | $\frac{1}{2}$ |
| $I = 0.9\text{ A}$                                              |               |



17. A battery of emf  $E$  and internal resistance  $r$  is connected to a rheostat. When a current of  $2A$  is drawn from the battery, the potential difference across the rheostat is  $5V$ . The potential difference becomes  $4V$  when a current of  $4A$  is drawn from the battery. Calculate the value of  $E$  and  $r$ .



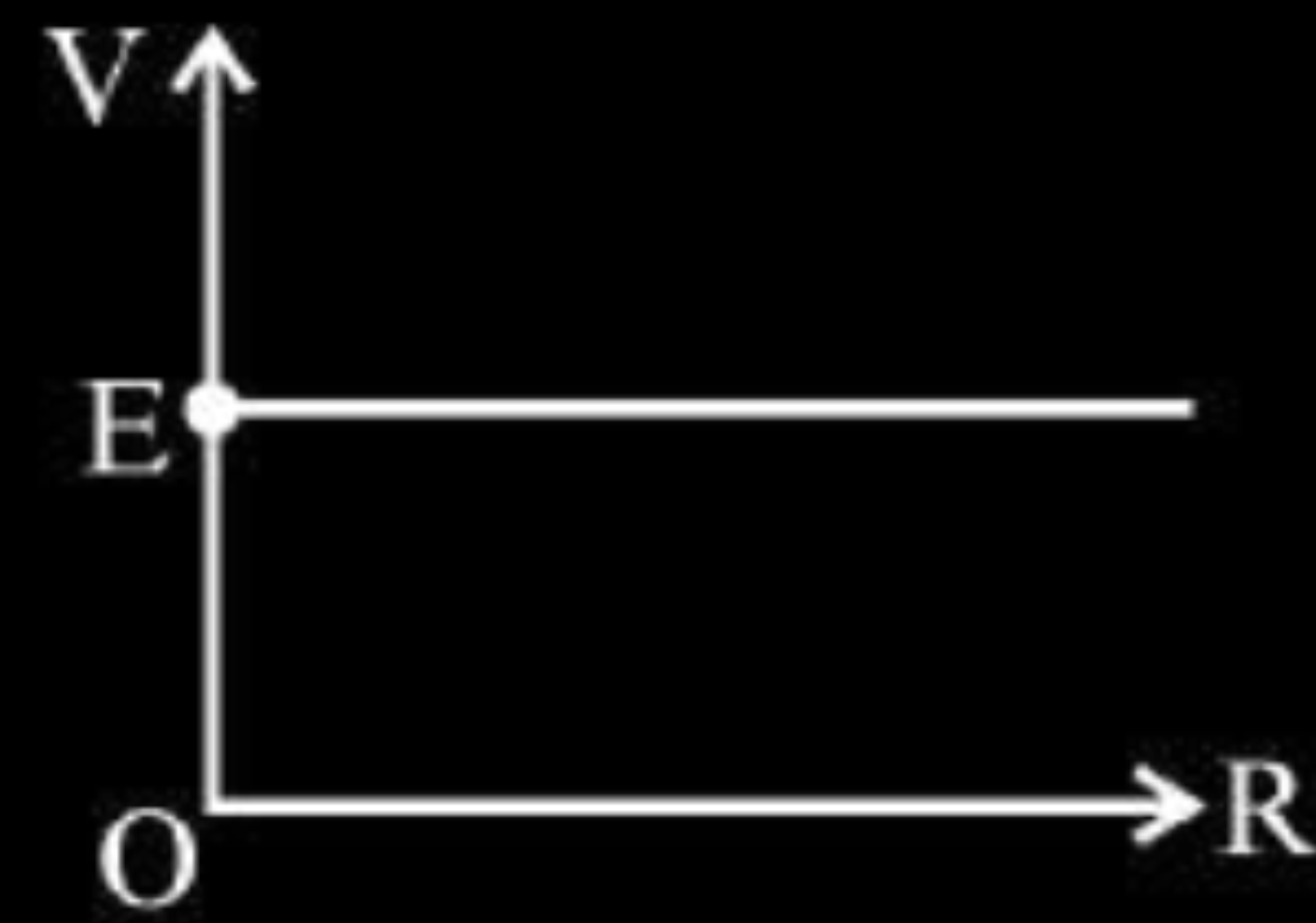
17. A battery of emf  $E$  and internal resistance  $r$  is connected to a rheostat. When a current of  $2A$  is drawn from the battery, the potential difference across the rheostat is  $5V$ . The potential difference becomes  $4V$  when a current of  $4A$  is drawn from the battery. Calculate the value of  $E$  and  $r$ .

2

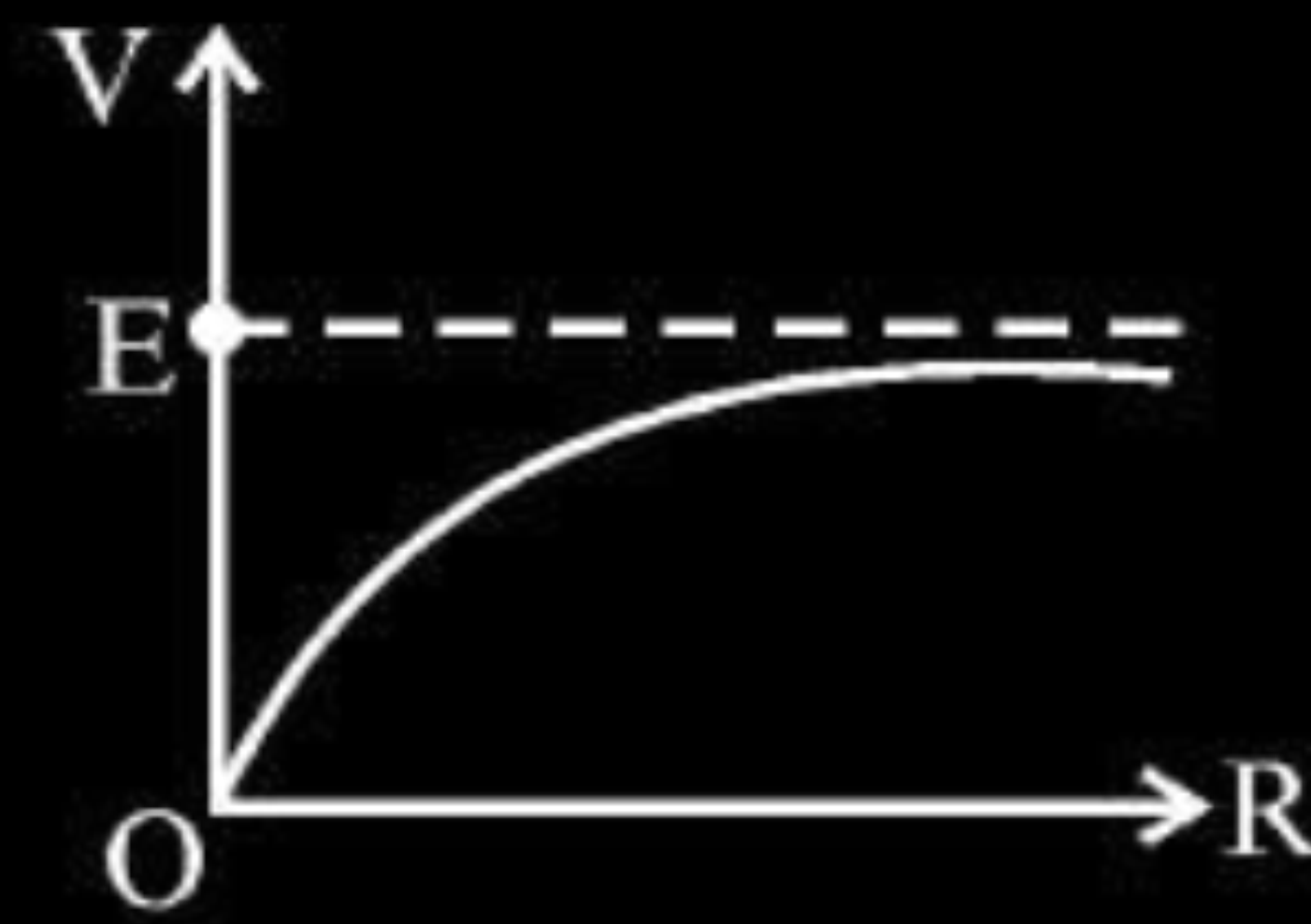
|                              |               |
|------------------------------|---------------|
| Calculating the value of $E$ | 1             |
| Calculating the value of $r$ | 1             |
| $E = V + Ir$                 |               |
| In first case                | $\frac{1}{2}$ |
| $E = 5 + 2r$                 |               |
| In second case               |               |
| $E = 4 + 4r$                 | $\frac{1}{2}$ |
| After solving                |               |
| $E = 6V$                     | $\frac{1}{2}$ |
| $r = 0.5\Omega$              | $\frac{1}{2}$ |



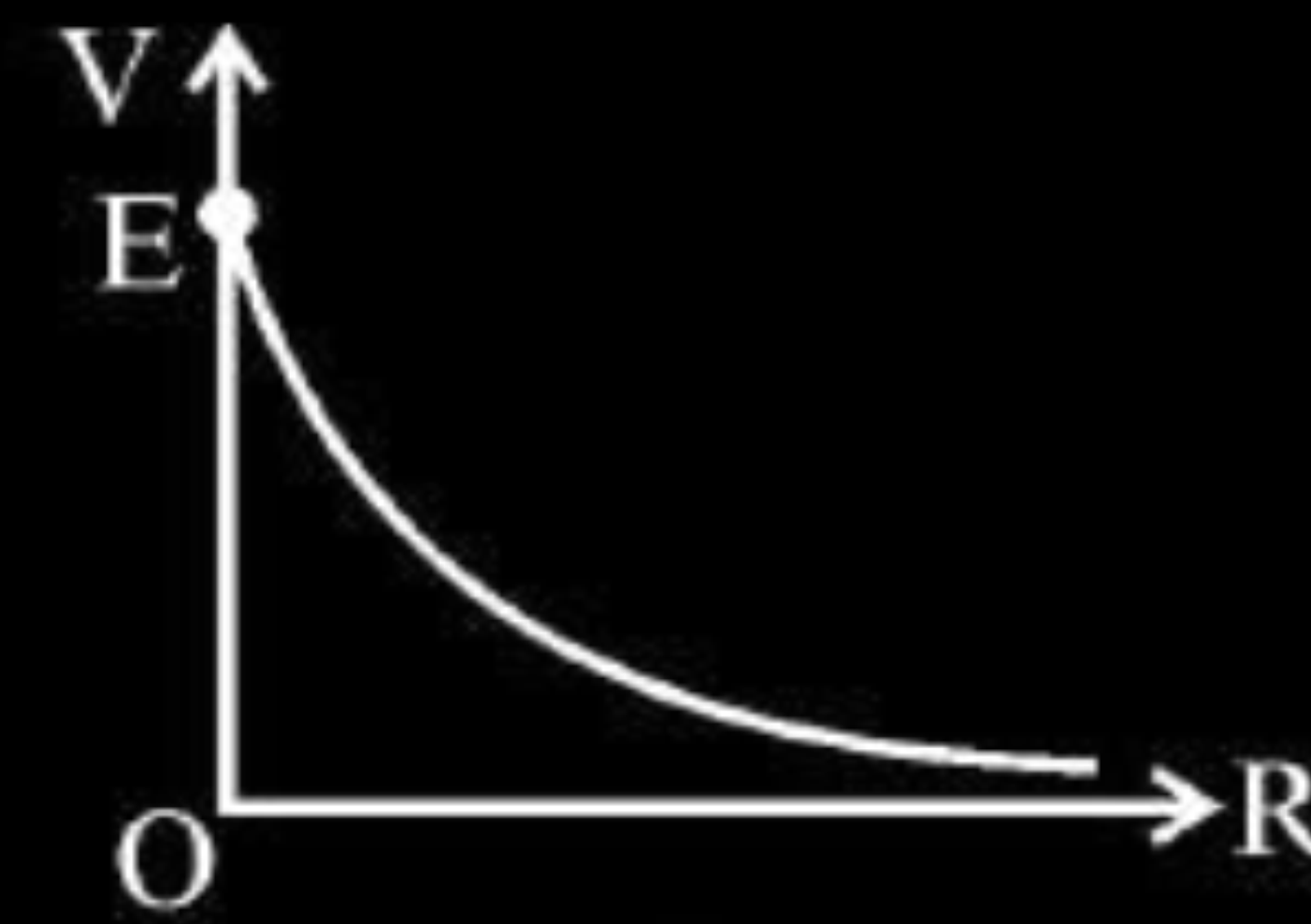
1. A cell of emf ( $E$ ) and internal resistance  $r$  is connected across a variable external resistance  $R$ . The graph of terminal potential difference  $V$  as a function of  $R$  is –



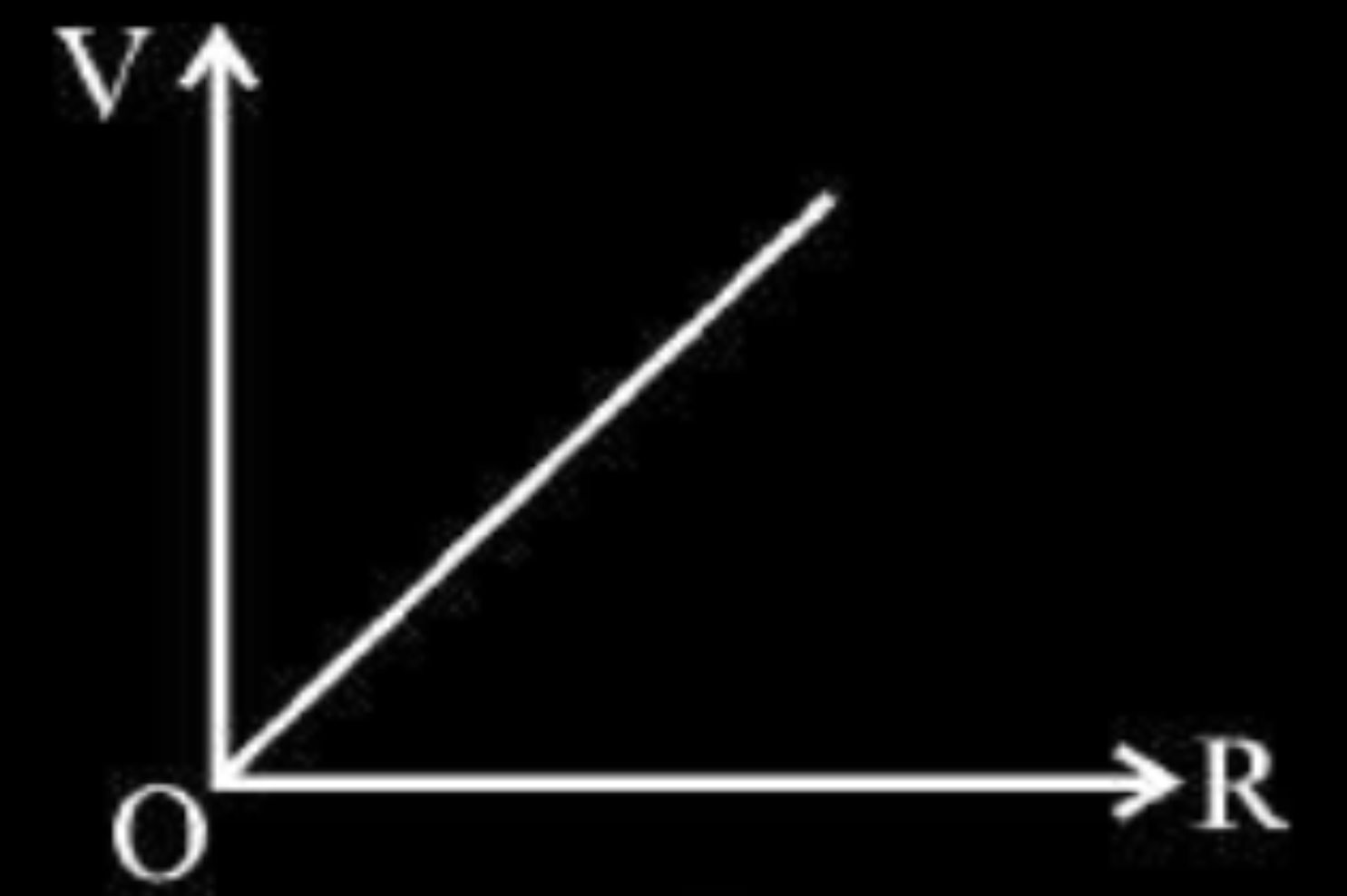
(a)



(b)



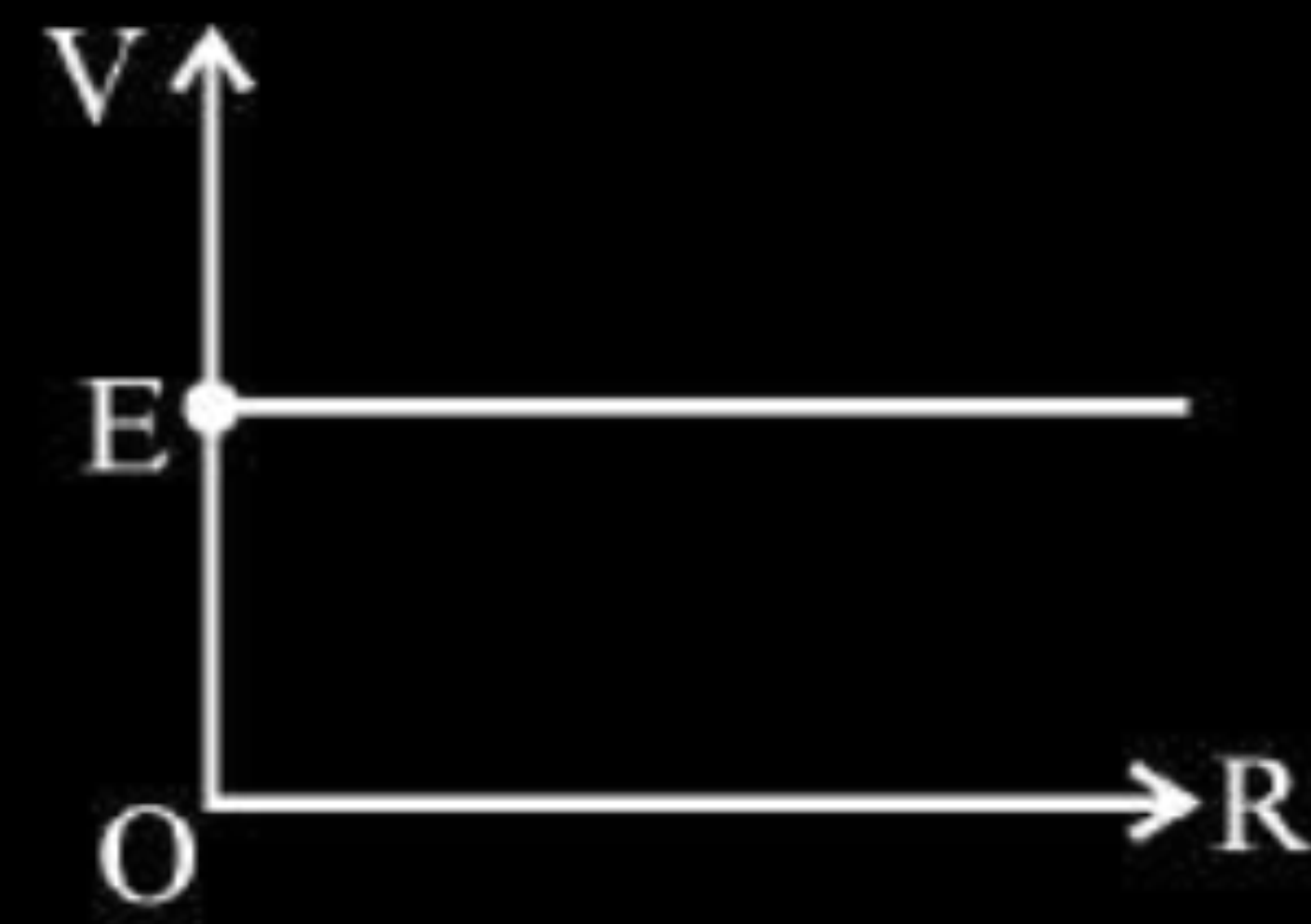
(c)



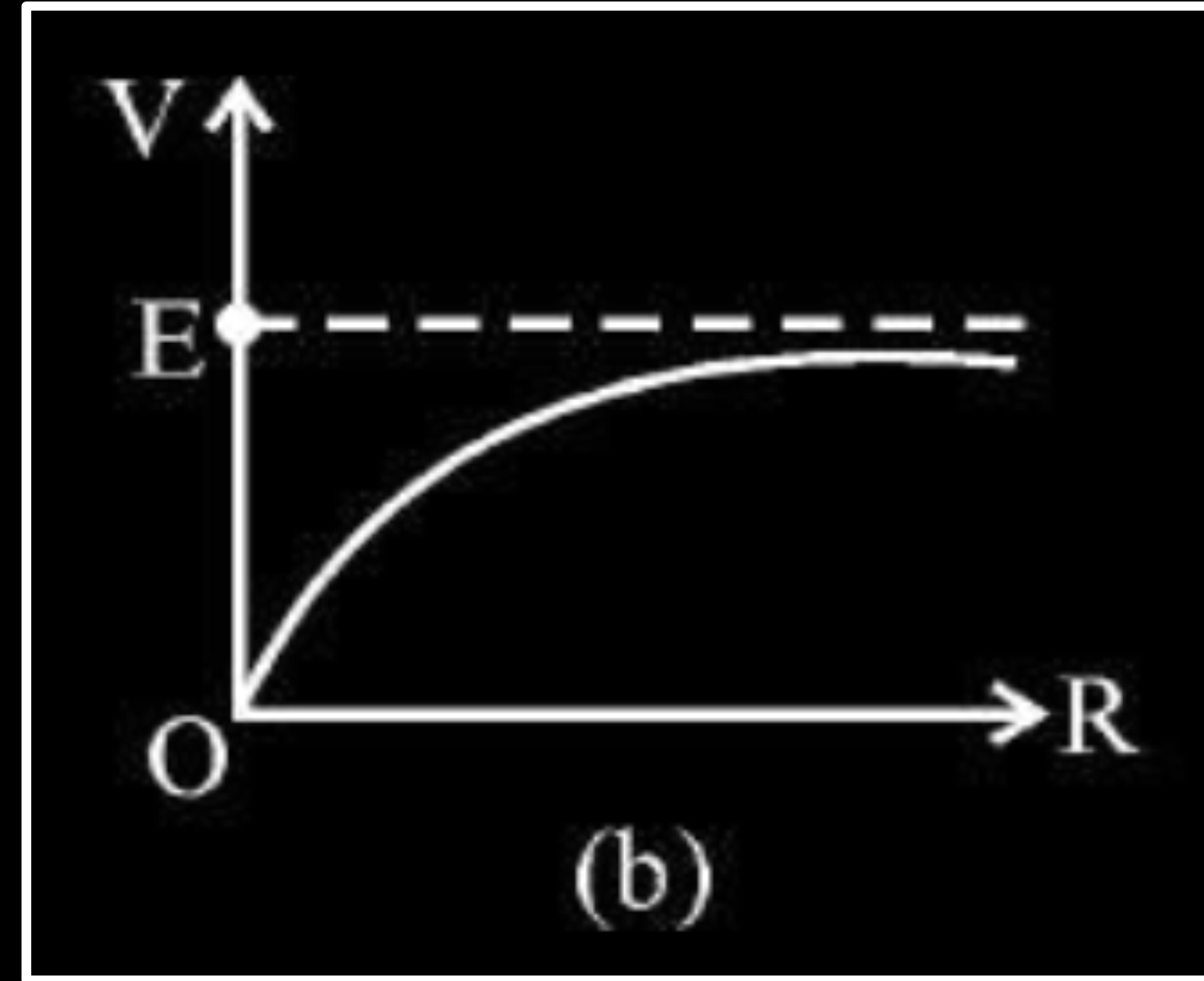
(d)



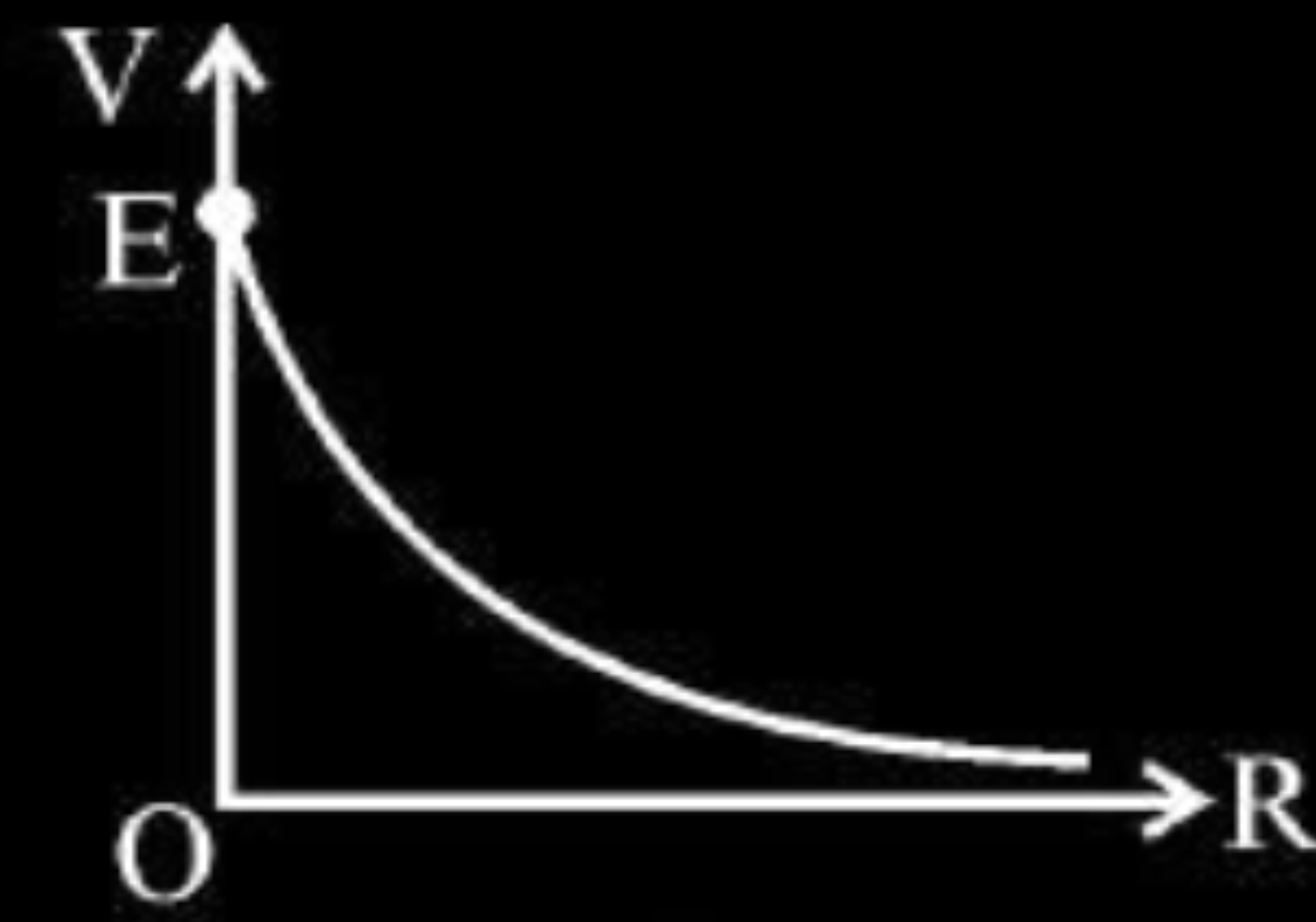
1. A cell of emf ( $E$ ) and internal resistance  $r$  is connected across a variable external resistance  $R$ . The graph of terminal potential difference  $V$  as a function of  $R$  is –



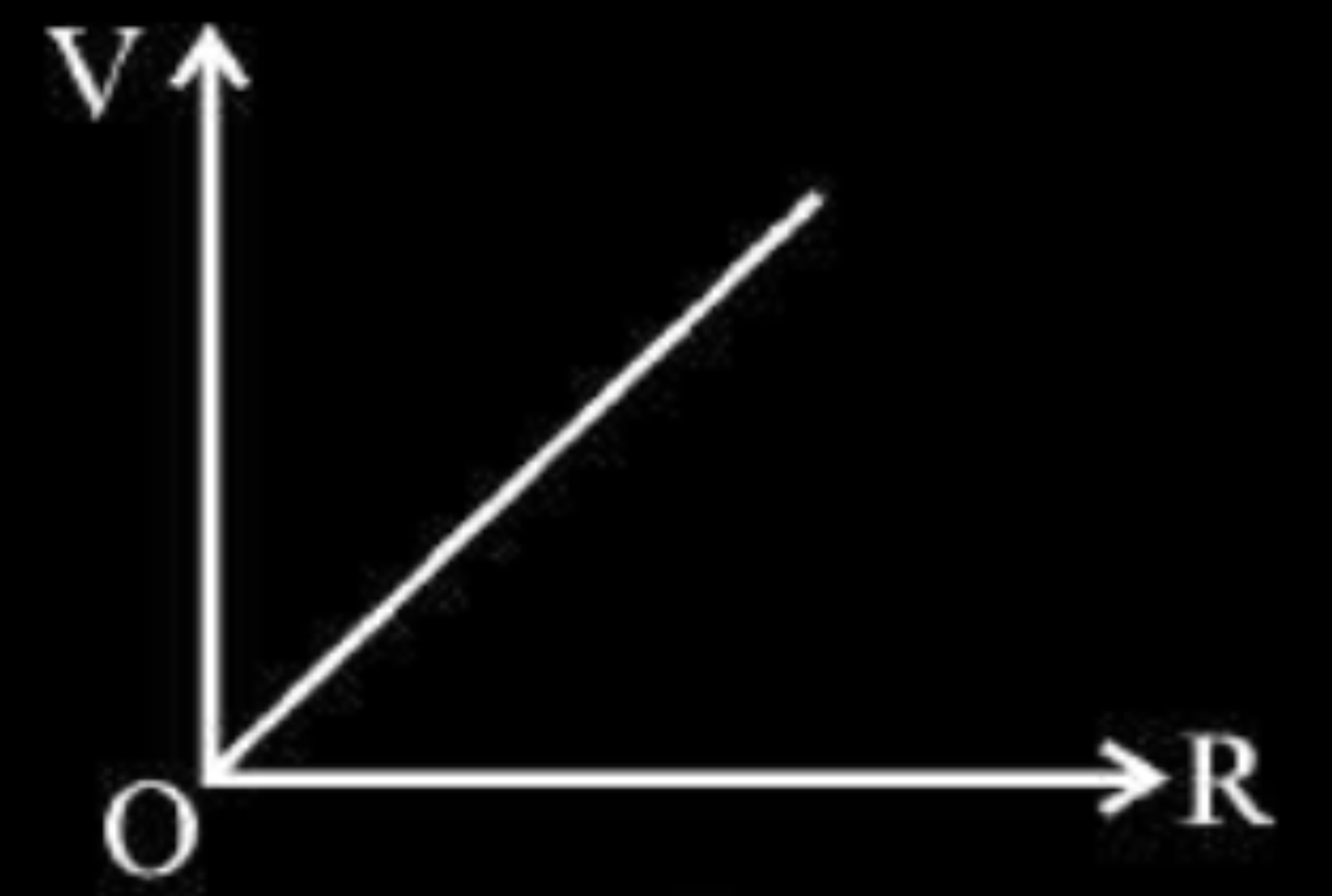
(a)



(b)



(c)



(d)



1. A cell of internal resistance  $r$  connected across an external resistance  $R$  can supply maximum current when

1

(A)  $R = r$

(B)  $R > r$

(C)  $R = \frac{r}{2}$

(D)  $R = 0$



1. A cell of internal resistance  $r$  connected across an external resistance  $R$  can supply maximum current when

1

(A)  $R = r$

(B)  $R > r$

(C)  $R = \frac{r}{2}$

(D)  $R = 0$



2. In a current carrying conductor, the ratio of the electric field and the current density at a point is called *1*

- (A) Resistivity
- (B) Conductivity
- (C) Resistance
- (D) Mobility



- 28.** (a) Two cells of emf  $E_1$  and  $E_2$  have their internal resistances  $r_1$  and  $r_2$ , respectively. Deduce an expression for the equivalent emf and internal resistance of their parallel combination when connected across an external resistance  $R$ . Assume that the two cells are supporting each other.
- (b) In case the two cells are identical, each of emf  $E = 5 \text{ V}$  and internal resistance  $r = 2 \Omega$ , calculate the voltage across the external resistance  $R = 10 \Omega$ .

3

|                        |          |
|------------------------|----------|
| a) Internal resistance | 1 ½ mark |
| b) Voltage across R    | 1 ½ mark |



Current drawn from cell -1

$$I_1 = \frac{E_1 - V}{r_1}$$

Current drawn from cell -2

$$I_2 = \frac{E_2 - V}{r_2}$$

Resultant current

$$I = I_1 + I_2$$

On solving

$$\therefore I = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2} - V \left( \frac{r_2 + r_1}{r_1 r_2} \right)$$

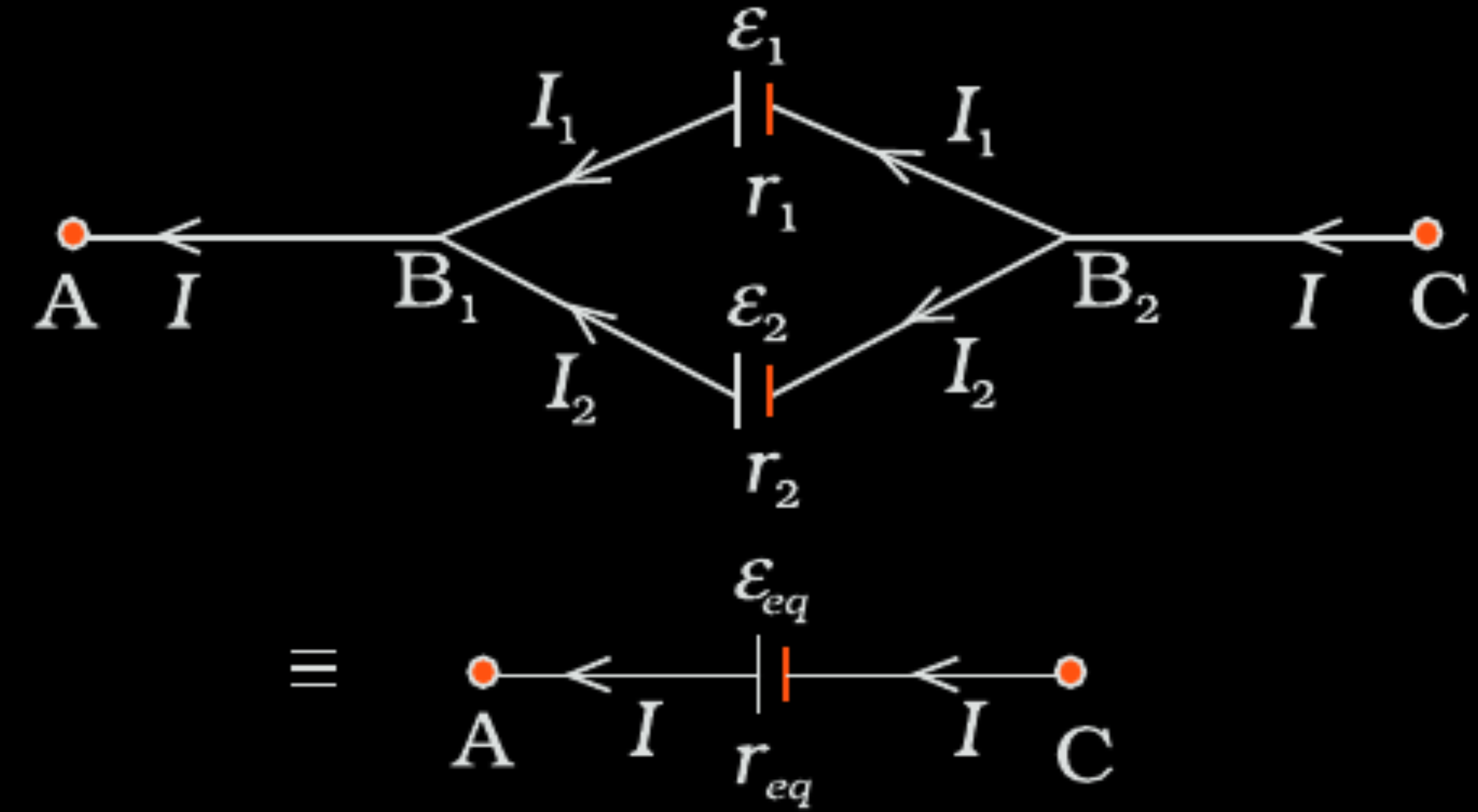
$$\therefore V = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2} - I \left( \frac{r_1 r_2}{r_2 + r_1} \right)$$

$$V = E_{eq} - I r_{eq}$$

$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_2 + r_1}$$

1/2



$$r_{eff} = \frac{r_1 r_2}{r_1 + r_2} = \frac{2 \times 2}{2 + 2} = 1 \Omega$$

Current through R

$$I = \frac{E_{effect}}{R + r_{eff}} = \frac{5}{10 + 1} = \frac{5}{11} A$$

P.D across R

$$= \frac{5}{11} \times 10 = 4.54 \text{ volt}$$

1/2

1/2

1/2

1/2

1/2

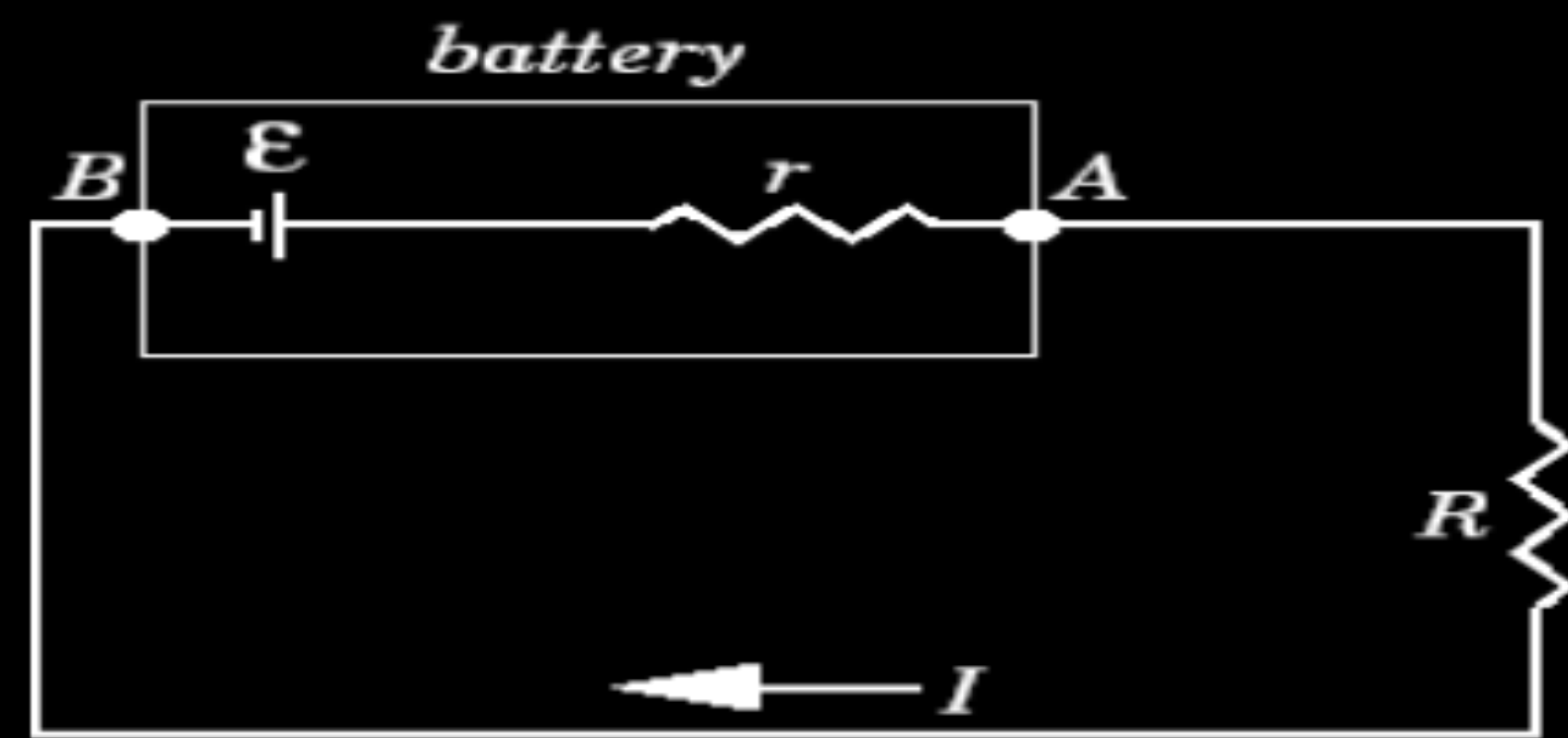


- (a) Derive a relation between the internal resistance, emf and terminal potential difference of a cell from which current  $I$  is drawn. Draw  $V$  vs  $I$  graph for a cell and explain its significance.
- (b) A voltmeter of resistance  $998\ \Omega$  is connected across a cell of emf  $2\text{ V}$  and internal resistance  $2\ \Omega$ . Find the potential difference across the voltmeter and also across the terminals of the cell. Estimate the percentage error in the reading of the voltmeter.

|                                       |               |
|---------------------------------------|---------------|
| a) Relation between $E$ , $V$ , $r$   | 1             |
| Graph $V$ v/s $I$                     | 1             |
| Significance of graph                 | 1             |
| b) Current in voltmeter               | 1             |
| Potential difference across voltmeter | $\frac{1}{2}$ |
| Percentage Error                      | $\frac{1}{2}$ |



(a)

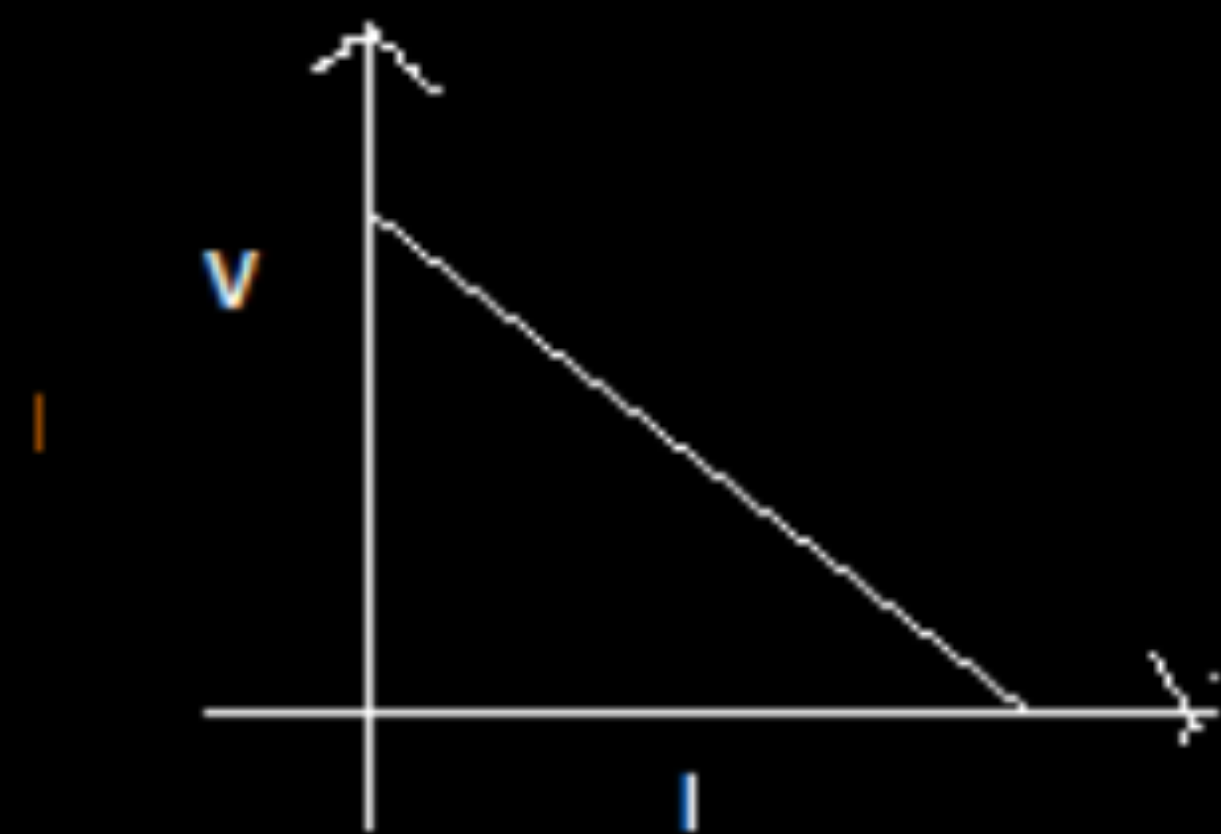


$$E - IR - rI = 0$$

$$E - V - Ir = 0$$

$$E = V + Ir$$

[Award 1 mark even if student writes the relation directly]



Significance of Graph – To find cmf and internal resistance of cell.

 $\frac{1}{2}$ 
 $\frac{1}{2}$ 

1

(b)

$$V = E - Ir$$

$$998 \times I = 2 - 2I$$

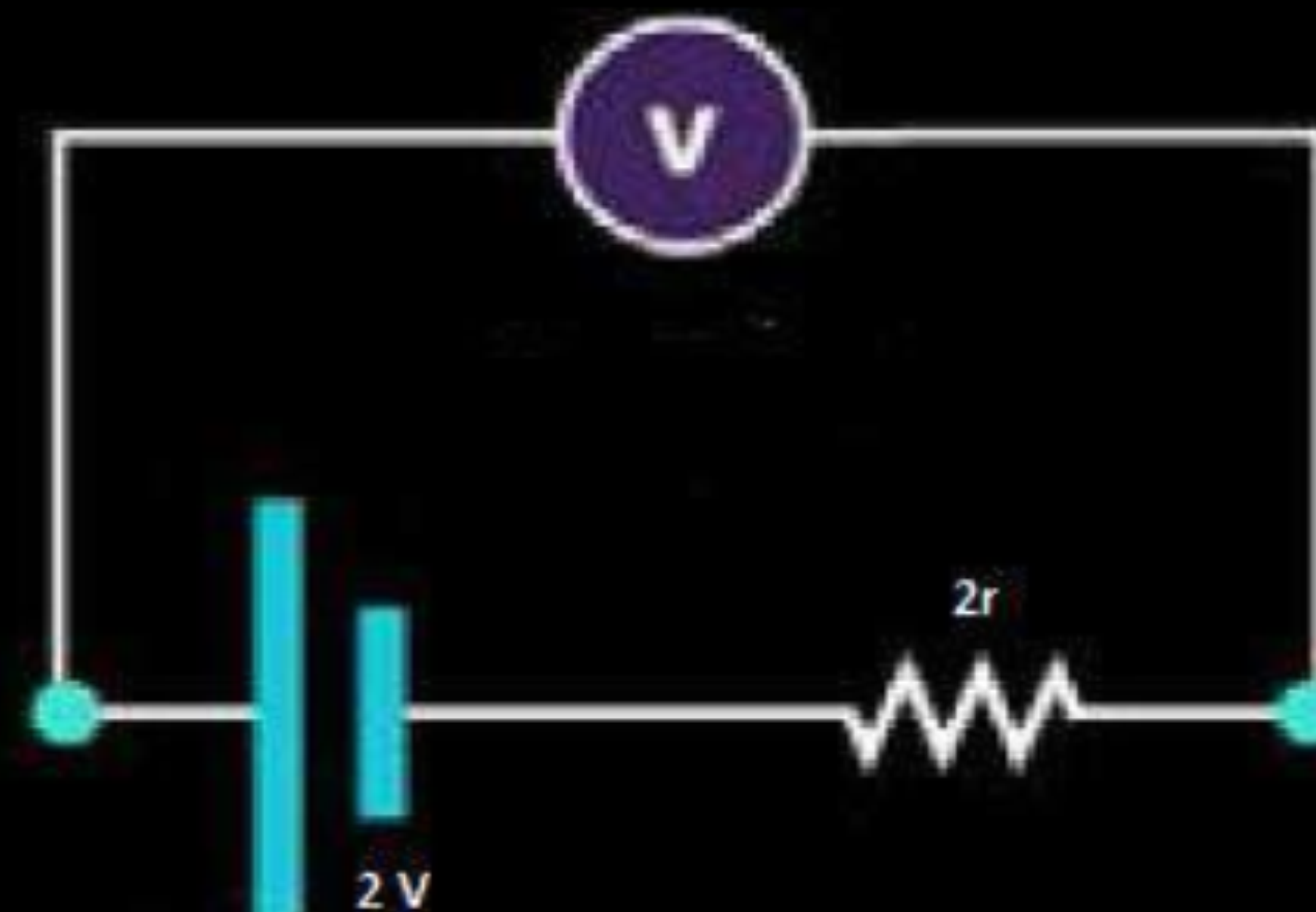
$$1000 \times I = 2$$

$$I = \frac{2}{1000} = .002 \text{ A}$$

$$V = .002 \times 998$$

$$V = 1.996 \text{ V}$$

$$\% \text{ error} = \frac{.004}{2} \times 100 = 0.2 \%$$


 $\frac{1}{2}$ 
 $\frac{1}{2}$ 
 $\frac{1}{2}$



- (a) Two cells of different emfs and internal resistances are connected in parallel with one another. Derive the expression for the equivalent emf and equivalent internal resistance of the combination.
- (b) Two identical cells of emf 1.5 V and internal resistance  $r$  are each connected in parallel providing a supply to an external circuit consisting of two resistances of  $17\ \Omega$  each joined in parallel. A very high resistance voltmeter reads the terminal voltage of the cell to be 1.4 V. Calculate the internal resistance of each cell.

|    |                                                               |                             |
|----|---------------------------------------------------------------|-----------------------------|
| a) | Expression of current in terms of $E$ and $V$                 | 1                           |
|    | Expression of voltage in terms of emf and internal resistance | 1                           |
|    | Expression for $E_{eq}$ and $r_{eq}$                          | $\frac{1}{2} + \frac{1}{2}$ |
| b) | Value of current & value of internal resistance               | 1+1                         |



$$I = I_1 + I_2$$

Potential difference across B<sub>1</sub> B<sub>2</sub>

$$V = E_1 - I_1 r_1 \qquad \Rightarrow \quad I_1 \quad = \quad \frac{E_1 - V}{r_1}$$

$$V = E_2 - I_2 r_2 \qquad \Rightarrow \quad I_2 \quad = \quad \frac{E_2 - V}{r_2}$$

$$I = I_1 + I_2$$

$$= \frac{E_1 - V}{r_1} + \frac{E_2 - V}{r_2}$$

$$= \big(\frac{E_1}{r_1} + \frac{E_2}{r_2}\big) \quad - \quad V \big(\frac{1}{r_1} + \frac{1}{r_2}\big)$$

$$V = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \quad - \quad \frac{I r_1 r_2}{r_1 + r_2}$$

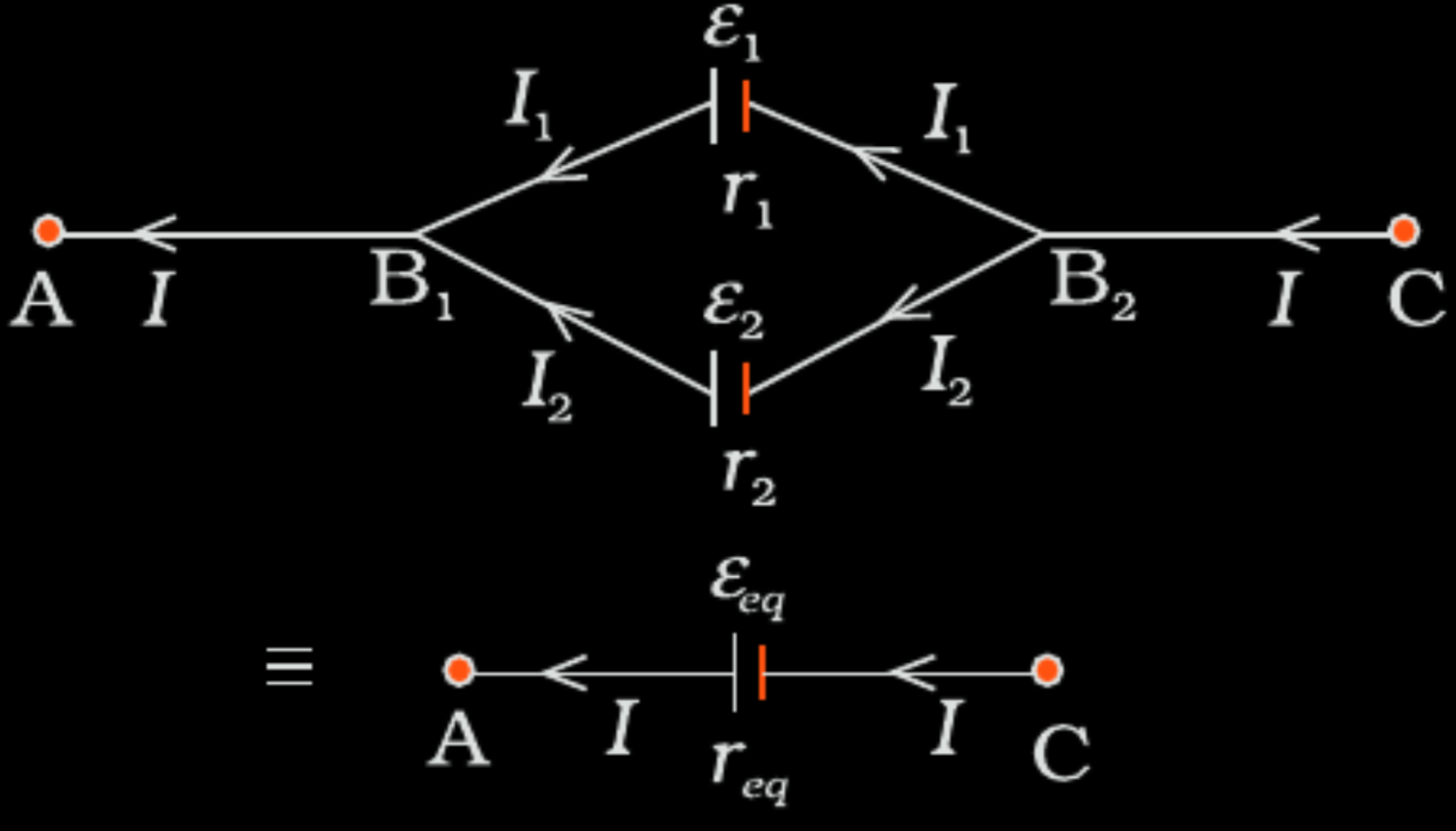
Compare with

$$V = E_{eq} - I r_{eg}$$

$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

$$\frac{1}{2}$$



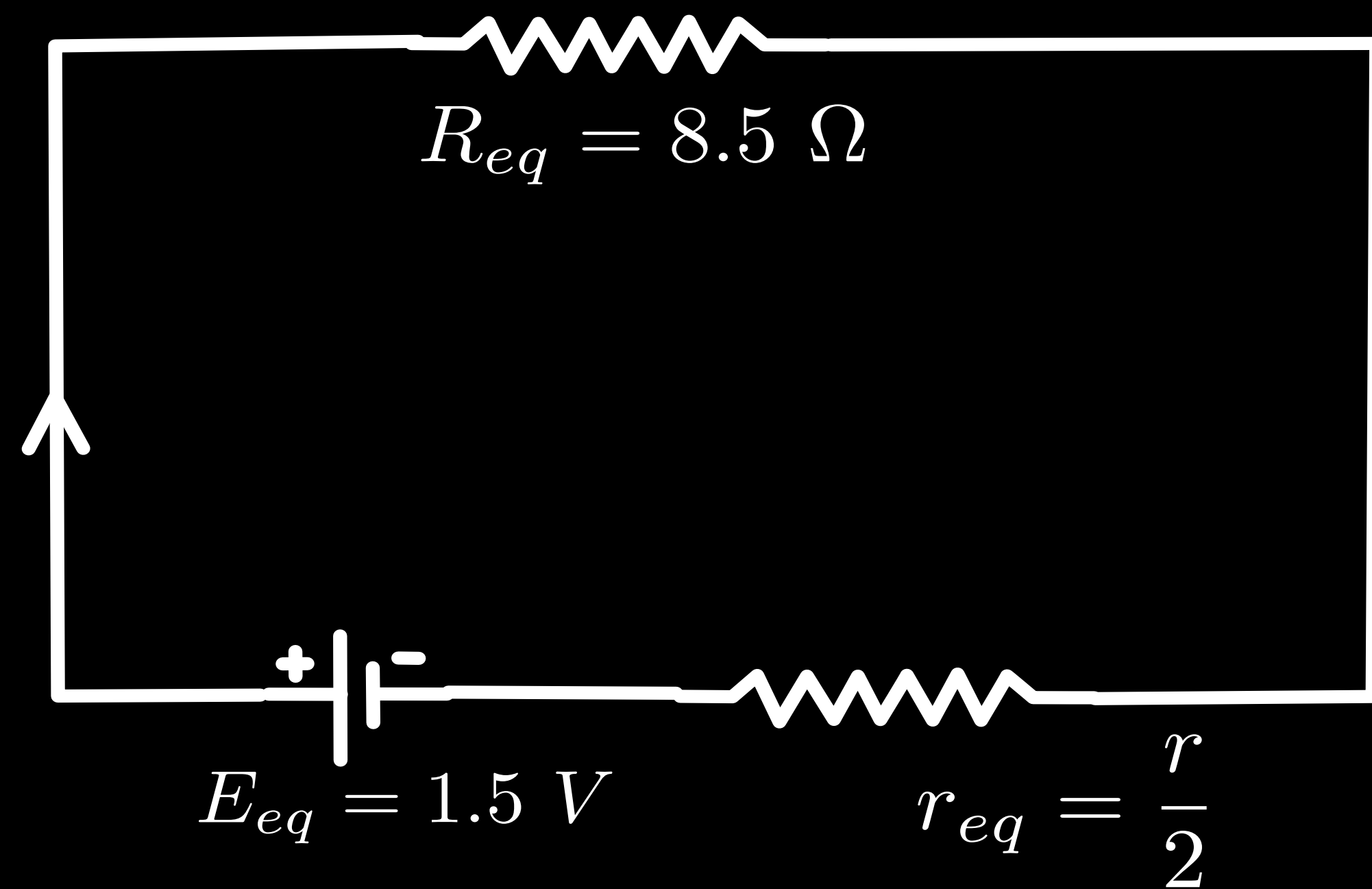
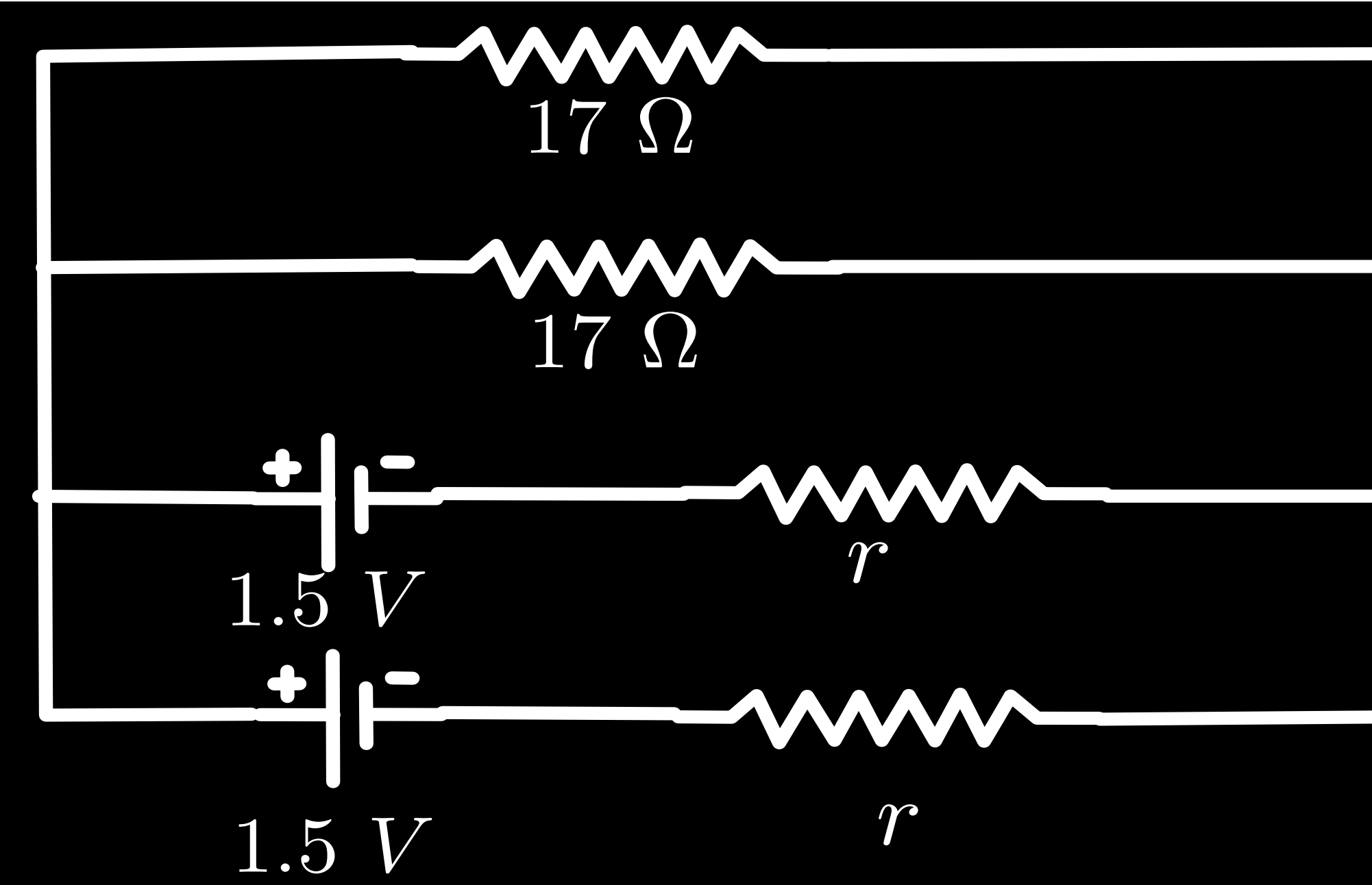
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$





Equivalent circuit

1

$$I = \frac{V}{R_{eq}} = \frac{1.4}{8.5}\text{ A}$$

And

$$V = E_{eq} - Ir_{eq}$$

$$\Rightarrow 1.4 = 1.5 - \frac{1.4}{8.5} \times \frac{r}{2}$$

$$\Rightarrow r = 1.21\Omega$$

1



Under what condition will the current in a wire be the same when connected in series and in parallel of  $n$  identical cells each having internal resistance  $r$  and external resistance  $R$  ?



Under what condition will the current in a wire be the same when connected in series and in parallel of  $n$  identical cells each having internal resistance  $r$  and external resistance  $R$  ?

|                                                              |   |
|--------------------------------------------------------------|---|
| When<br>$R + nr = nR + r$<br><b>Alternatively</b><br>$r = R$ | 1 |
|--------------------------------------------------------------|---|

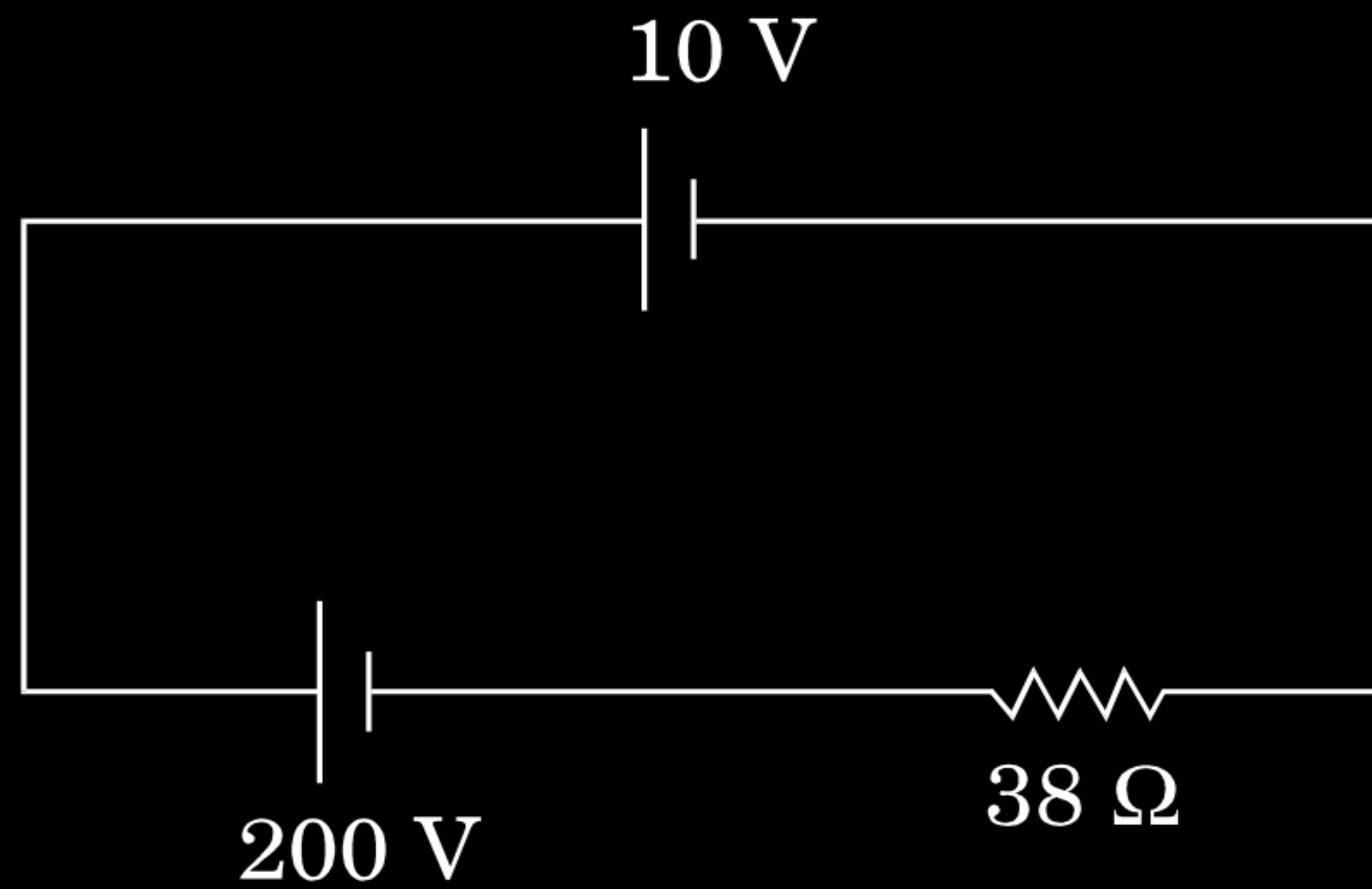


Two cells of emfs  $\varepsilon_1$  &  $\varepsilon_2$  and internal resistances  $r_1$  &  $r_2$  respectively are connected in parallel. Obtain expressions for the equivalent.

- (i) resistance and
- (ii) emf of the combination

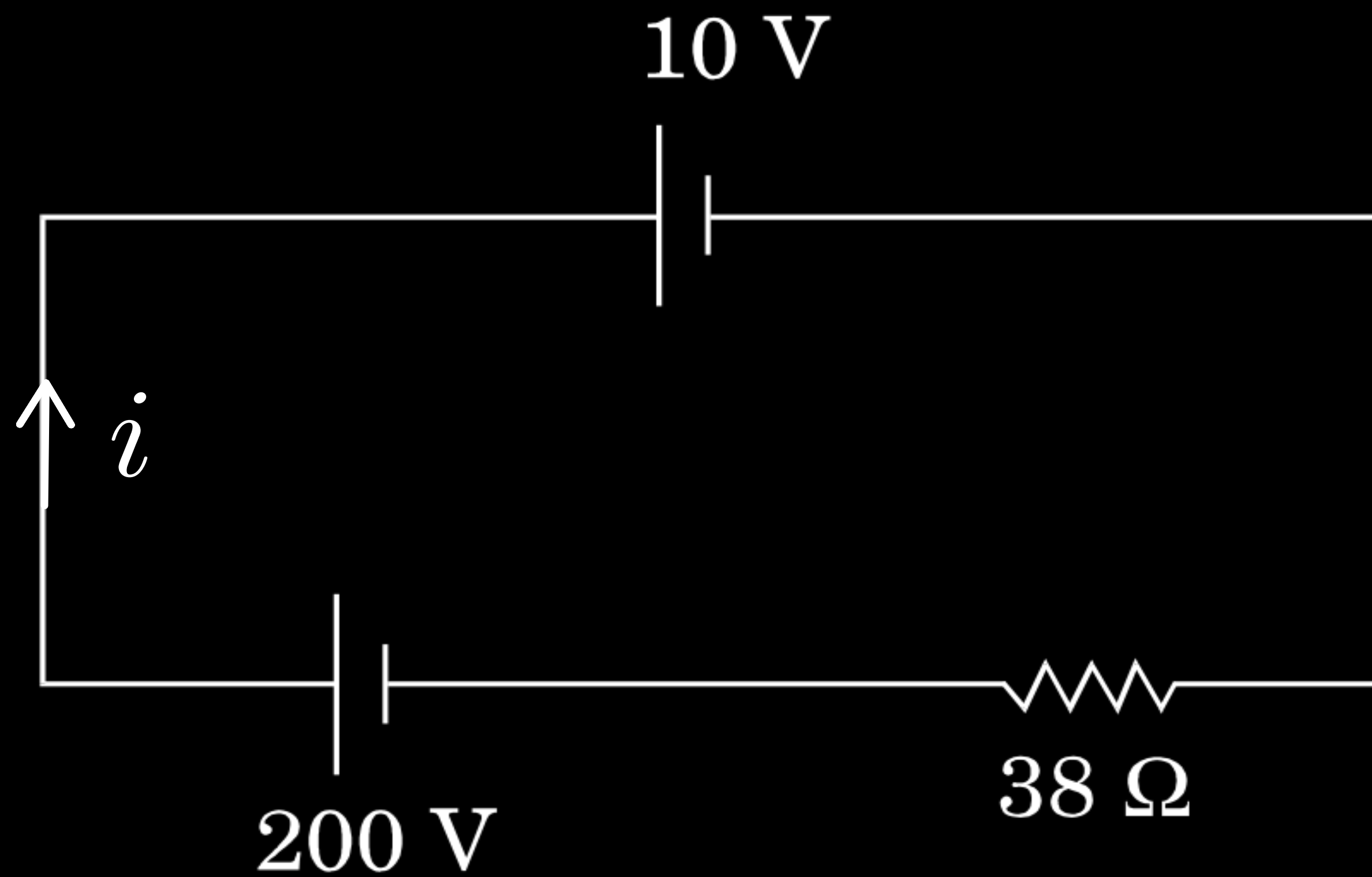


A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance  $38\ \Omega$  as shown in the figure. Find the value of current in the circuit.





A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance  $38\ \Omega$  as shown in the figure. Find the value of current in the circuit.



By Kirchhoff's law, we have, for the loop ABCD,

$$+200 - 38i - 10 = 0$$

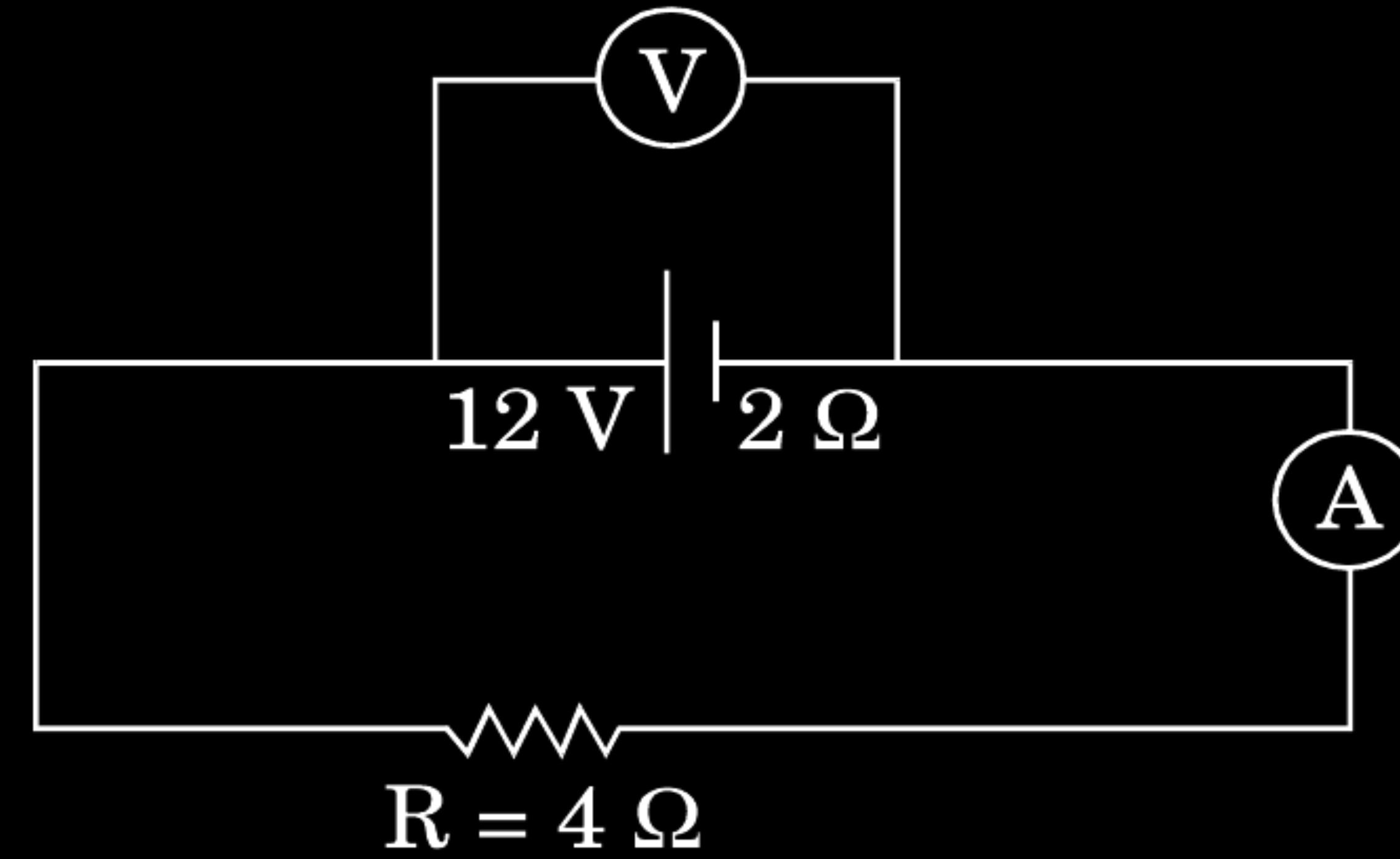
$$\therefore i = \frac{190}{38} \text{ A} = 5 \text{ A}$$

**1**

**1**

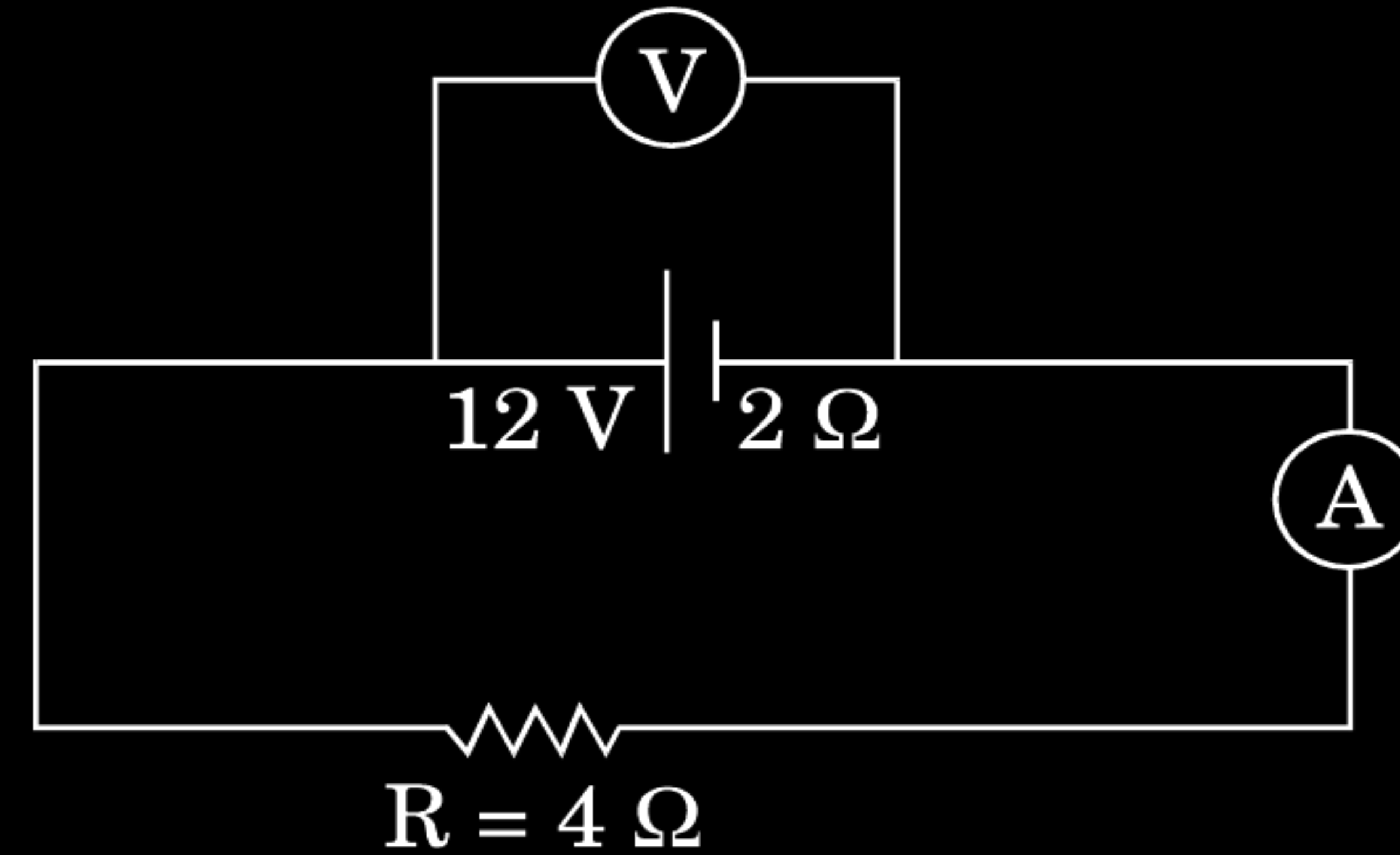


- (b) In the figure shown, an ammeter A and a resistor of  $4\ \Omega$  are connected to the terminals of the source. The emf of the source is  $12\text{ V}$  having an internal resistance of  $2\ \Omega$ . Calculate the voltmeter and ammeter readings.





- (b) In the figure shown, an ammeter A and a resistor of  $4\ \Omega$  are connected to the terminals of the source. The emf of the source is  $12\text{ V}$  having an internal resistance of  $2\ \Omega$ . Calculate the voltmeter and ammeter readings.



b) Ammeter Reading  $I = \frac{V}{R+r}$

$$= \frac{12}{4+2} \text{ A} = 2\text{ A}$$

Voltmeter Reading  $V = E - Ir$

$$= [12 - (2 \times 2)] \text{ V} = 8\text{ V}$$

(Alternatively,  $V = iR = 2 \times 4\text{ V} = 8\text{ V}$ )

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



Two cells of emfs  $1.5\text{ V}$  and  $2.0\text{ V}$  having internal resistances  $0.2\ \Omega$  and  $0.3\ \Omega$  respectively are connected in parallel. Calculate the emf and internal resistance of the equivalent cell.



Two cells of emfs 1.5 V and 2.0 V having internal resistances  $0.2\ \Omega$  and  $0.3\ \Omega$  respectively are connected in parallel. Calculate the emf and internal resistance of the equivalent cell.

$$\text{emf} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$= \frac{1.5 \times 0.3 + 2 \times 0.2}{0.2 + 0.3} \text{ V}$$

$$= \frac{0.45 + 0.40}{0.5} \text{ V} = 1.7 \text{ V}$$

$$r = \frac{r_1 r_2}{r_1 + r_2}$$

$$= \frac{0.2 \times 0.3}{0.2 + 0.3} \Omega$$

$$= \frac{0.06}{0.5} \Omega$$

$$= 0.12 \Omega$$

 $\frac{1}{2}$ 
 $\frac{1}{2}$ 
 $\frac{1}{2}$ 
 $\frac{1}{2}$



Two identical cells of emf 1.5 V each joined in parallel supply energy to an external circuit consisting of two resistances of  $7\ \Omega$  each joined in parallel. A very high resistance voltmeter reads the terminal voltage of cells to be 1.4 V. Calculate the internal resistance of each cell.



We have, for a single cell,

$$r = \left( \frac{E}{V} - 1 \right) R$$

$\therefore$  For the parallel combination, as given in the question,

$$\frac{r}{2} = \left( \frac{E}{V} - 1 \right) \frac{R}{2}$$

$$\therefore r = \left( \frac{1.5}{1.4} - 1 \right) \times 7 \Omega$$

$$= \frac{0.1}{1.4} \times 7 \Omega$$

$$= 0.5 \Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

**(Alternatively,**

$$I = \frac{V}{(R/2)}$$

$$\text{And } E = V - I(r/2)$$

$$\text{This gives } I = \frac{1.4}{7/2} \text{ A} = 0.4 \text{ A}$$

$$\therefore \frac{r}{2} = \frac{1.5 - 1.4}{0.4} = 0.25$$

$$\therefore r = 0.5 \Omega$$

( Note: If the student just draws the circuit diagram of the setup but does not do any calculations, award 1 mark only.)

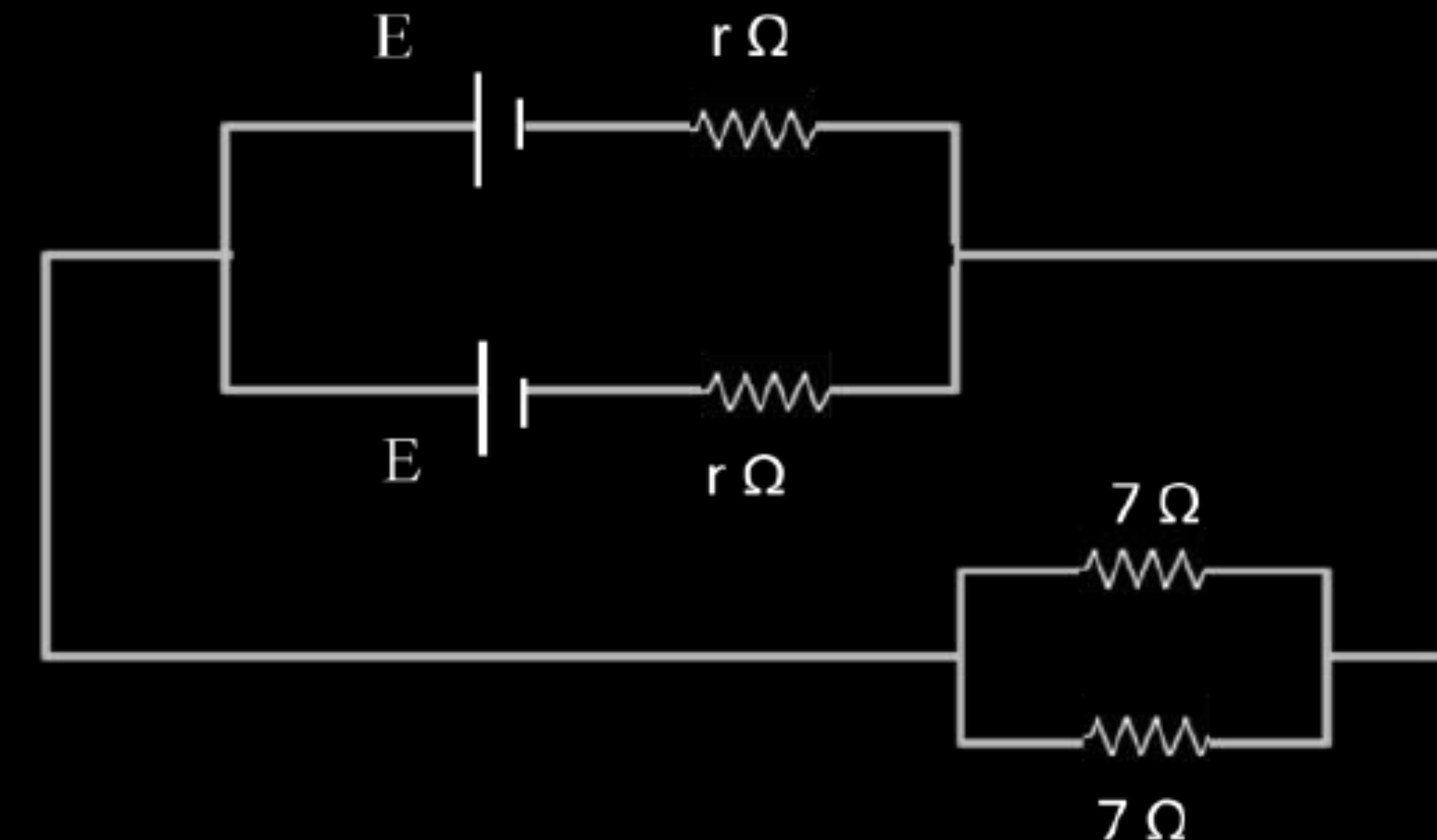
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

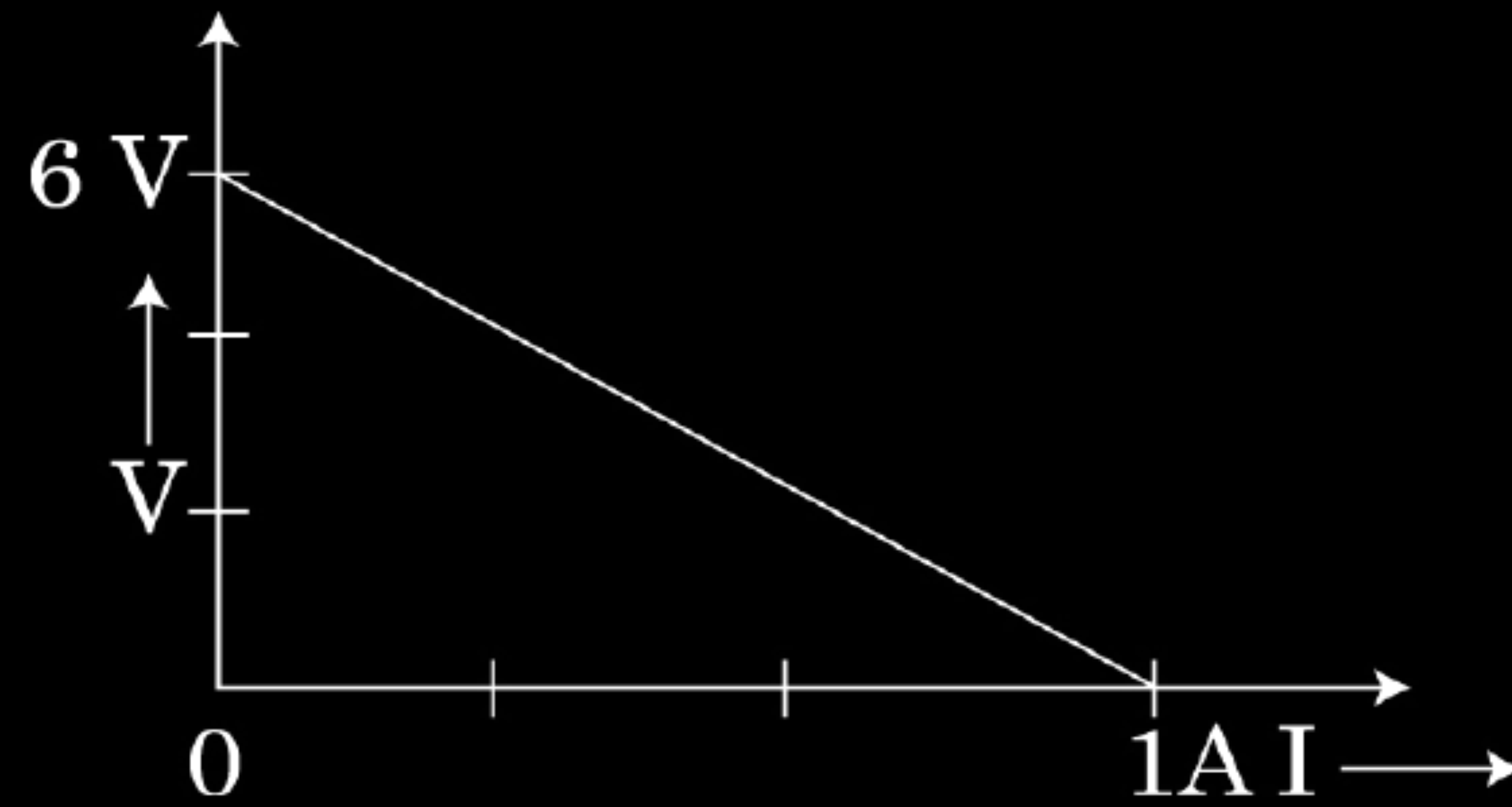
$\frac{1}{2}$

$\frac{1}{2}$



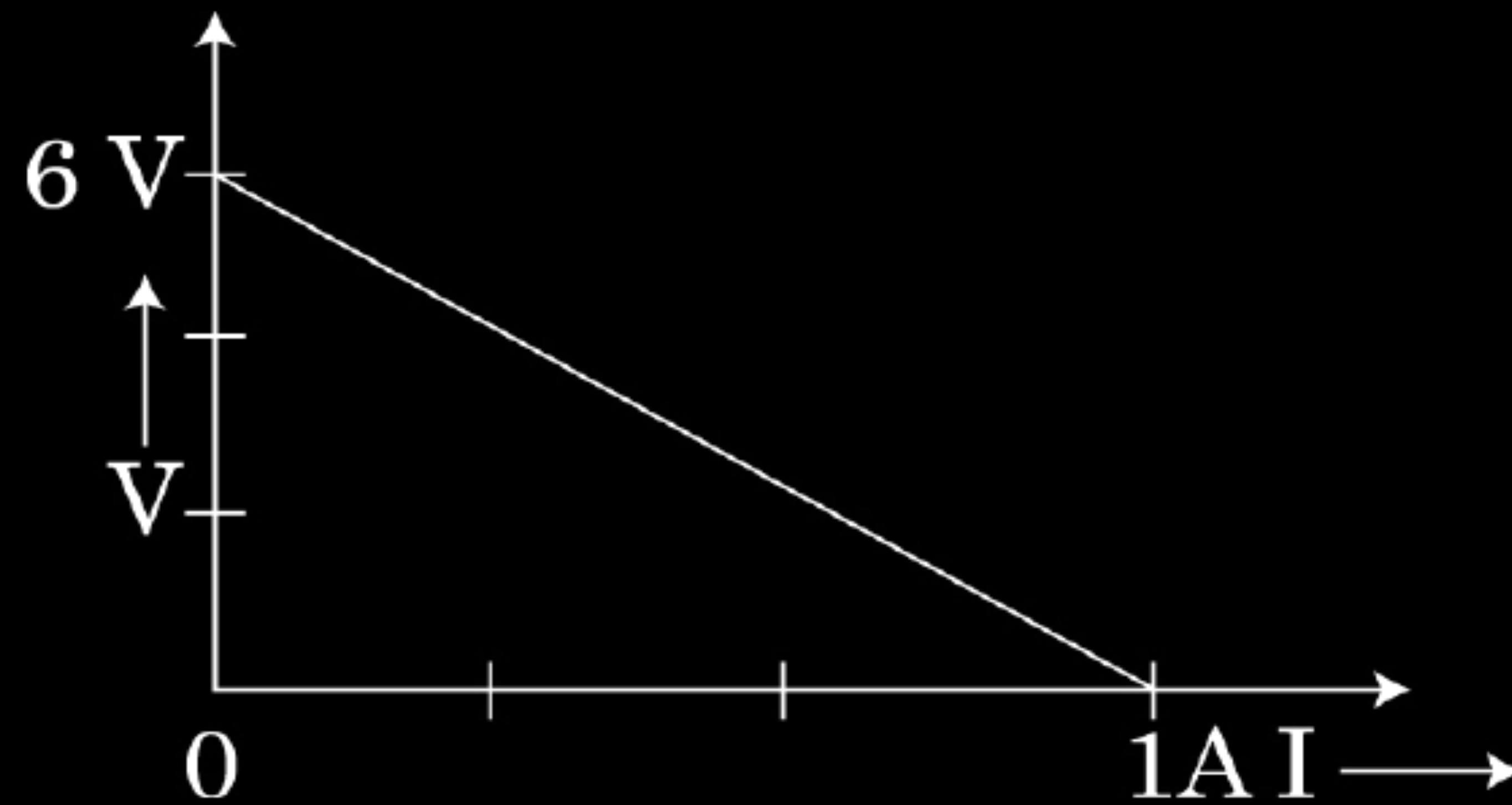


The plot of the variation of potential difference across a combination of three identical cells in series, versus current is shown below. What is the emf and internal resistance of each cell ?





The plot of the variation of potential difference across a combination of three identical cells in series, versus current is shown below. What is the emf and internal resistance of each cell ?

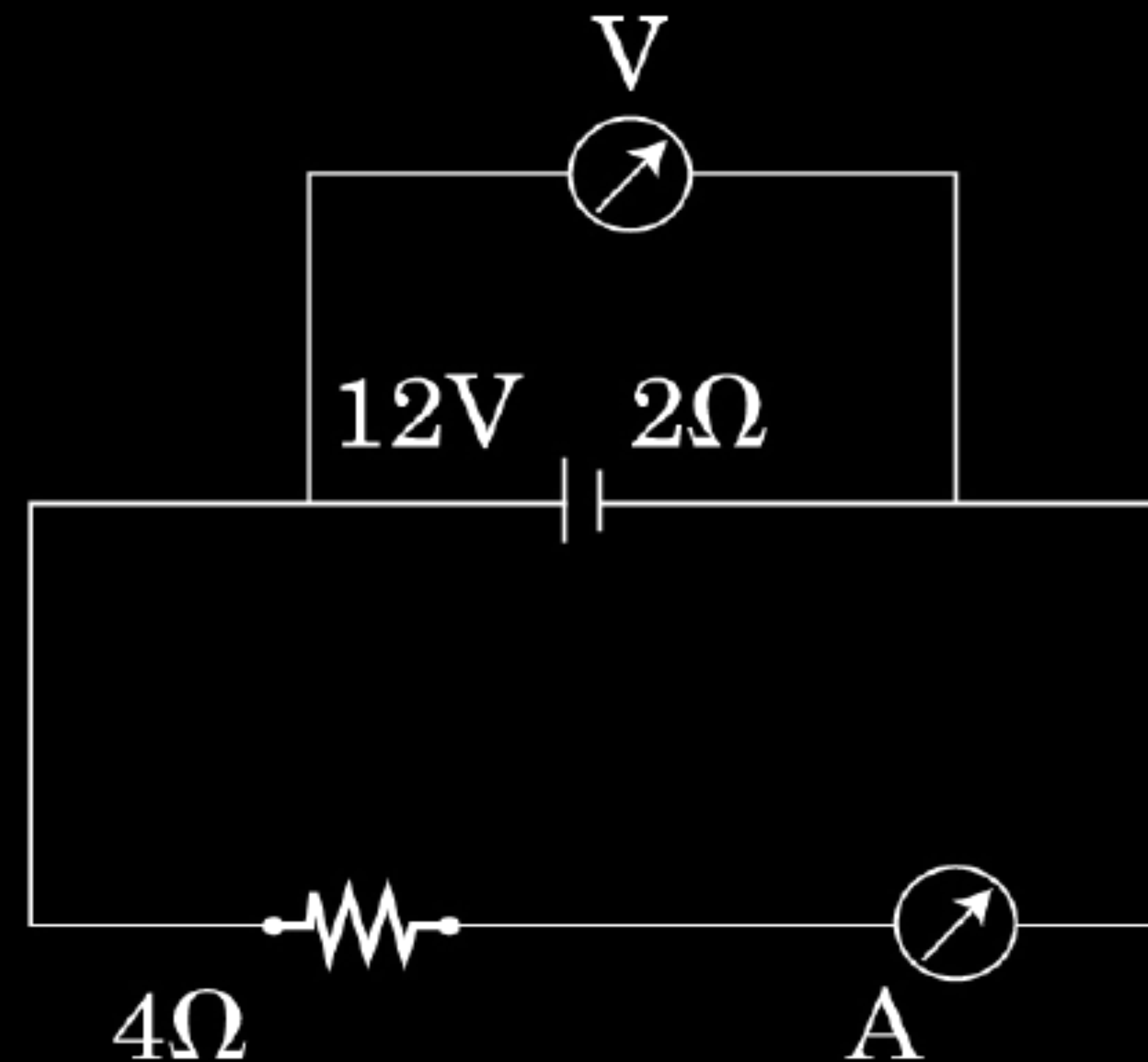


|                 |               |
|-----------------|---------------|
| $E = 2\text{V}$ | $\frac{1}{2}$ |
| $r = 2\Omega$   | $\frac{1}{2}$ |



A battery of emf  $12\text{V}$  and internal resistance  $2\ \Omega$  is connected to a  $4\ \Omega$  resistor as shown in the figure.

- (a) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.
- (b) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit ?





a) Effective resistance of the circuit  $R_E = 6\Omega$

$$\therefore I = \frac{12A}{6} = 2A$$

Terminal potential difference across the cell,  $V = E - ir$

$\frac{1}{2}$

Also p.d. across  $4\Omega$  resistor  $= 4 \times 2V = 8V$

Hence the voltmeter gives the same reading in the two cases.

$\frac{1}{2}$

b) In series -current same

$\frac{1}{2}$

In parallel – potential same

$\frac{1}{2}$